

Lecture 5-2: Introduction to quantum mechanics-1

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/10/15

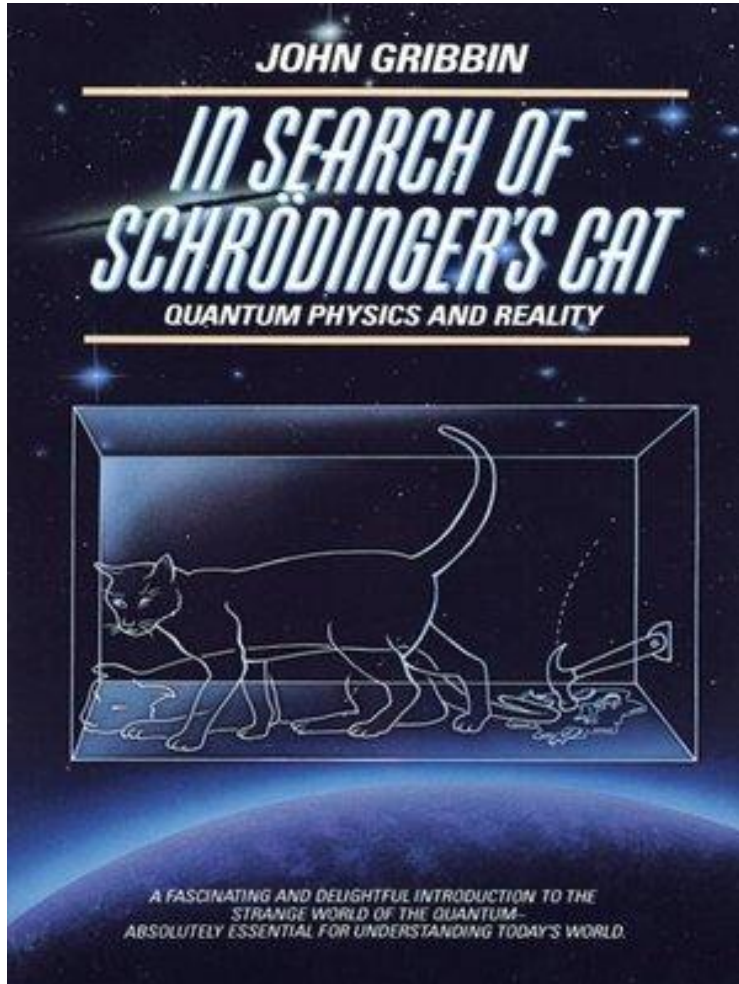
Major points of today's lecture

- **Introduction to quantum mechanics**

1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. The Bohr model: predicting discrete energy levels in atoms
4. Evidence for wave-particle duality
5. The Schrödinger equation
6. Quantum mechanics of particle-in-a-box models
7. Wave functions for particles in two- and three-dimensional boxes

Introduction to quantum mechanics

薛定谔的猫、麦克斯韦妖、芝诺的乌龟、拉普拉斯兽



“I don’t like it, and I’m sorry I ever had anything to do with it.”

Erwin Schrödinger, 1887-1961

“in search of Schrödinger's cat” by John Gribbin

什么是量子力学？

Original 长尾君 长尾科技 2022-01-27 19:47

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#主线文章 12 #量子力学 2

Introduction to quantum mechanics (old)

One of the fundamental assumptions of early science was that nature is continuous; that is, nature does not make “jumps.”



The central idea of quantum theory is that energy, like matter, is not continuous but it exists only in discrete packets.

The word quantum is the neuter singular of the Latin interrogative adjective quantus, meaning "how much".

quantum

noun [C] • PHYSICS • specialized

UK  /'kwɒn.təm/ US  /'kwɑːn.təm/

plural **quanta**

the smallest amount or unit of something, especially energy:

► **mechanics** [U]

the study of the effect of physical forces on objects and their movement

Major points of today's lecture

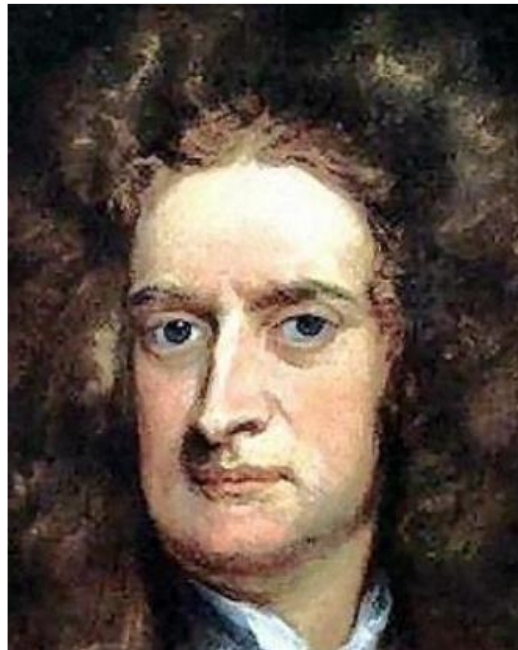
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Preliminaries: light (particle?)

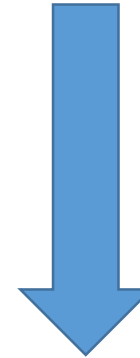


**Great Plague of London
1665-1666**



**Isaac Newton
(1643-1727)**

**Classical mechanics
Newton's second law: $F = ma$**



Initial status:

x_0, y_0, z_0
 v_0, p_0, E_0

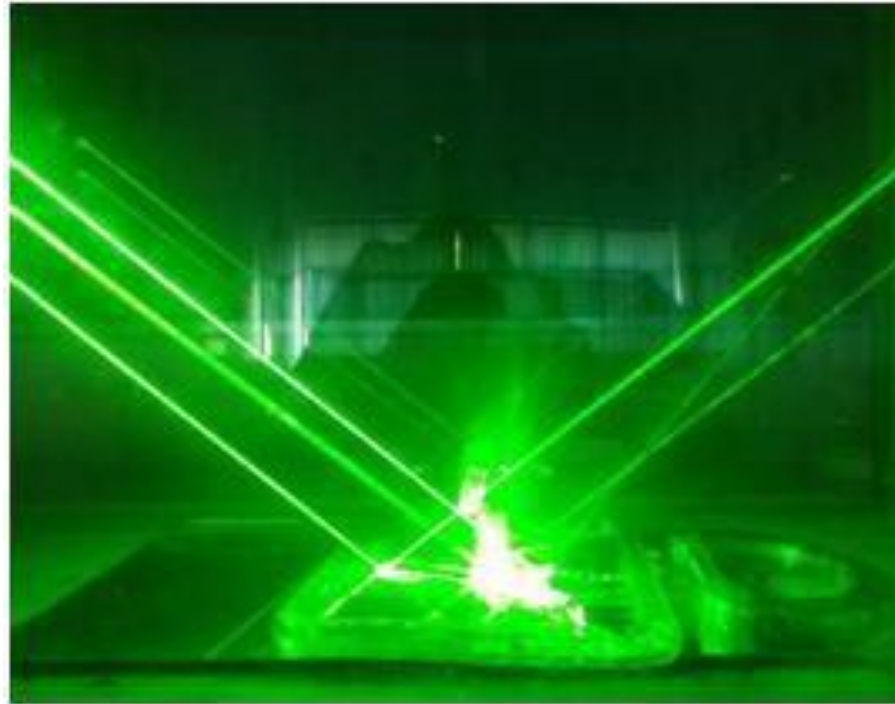
Final status:

x, y, z
 v, p, E

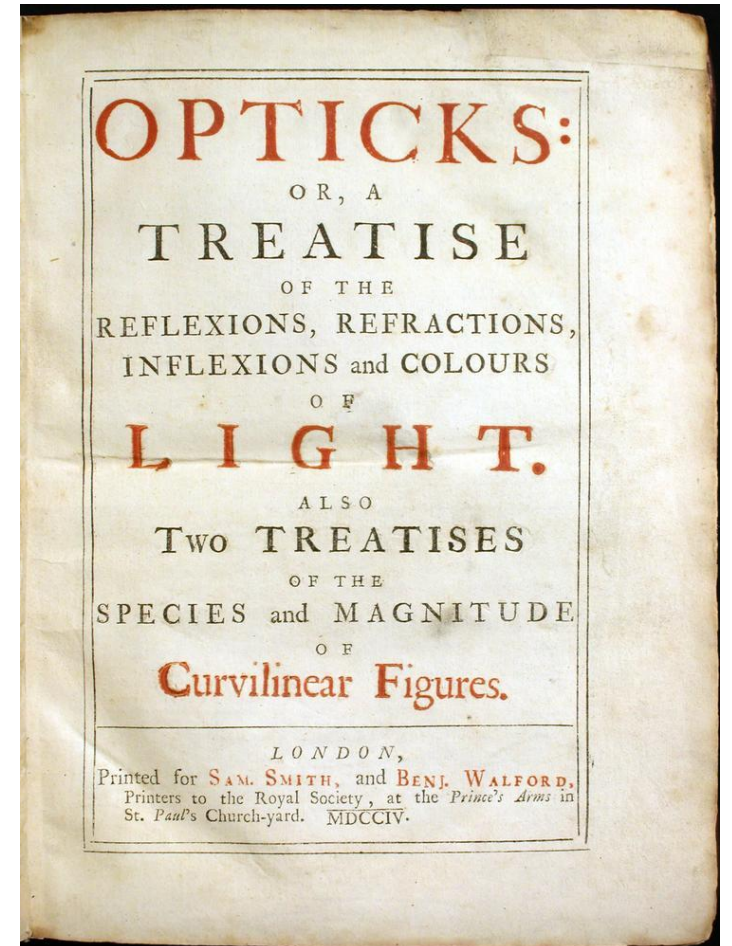
Preliminaries: light (particle?)



Color Spectrum
1666



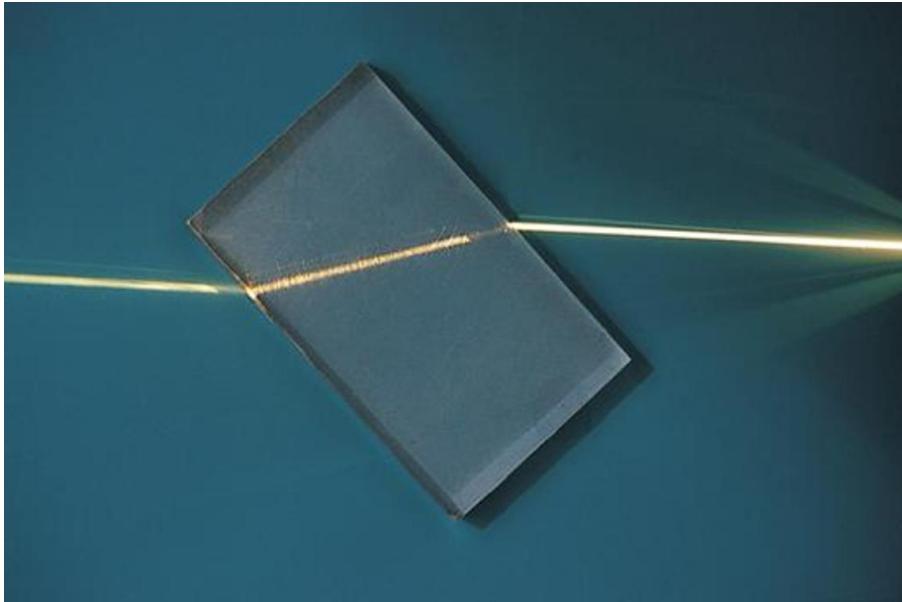
reflection



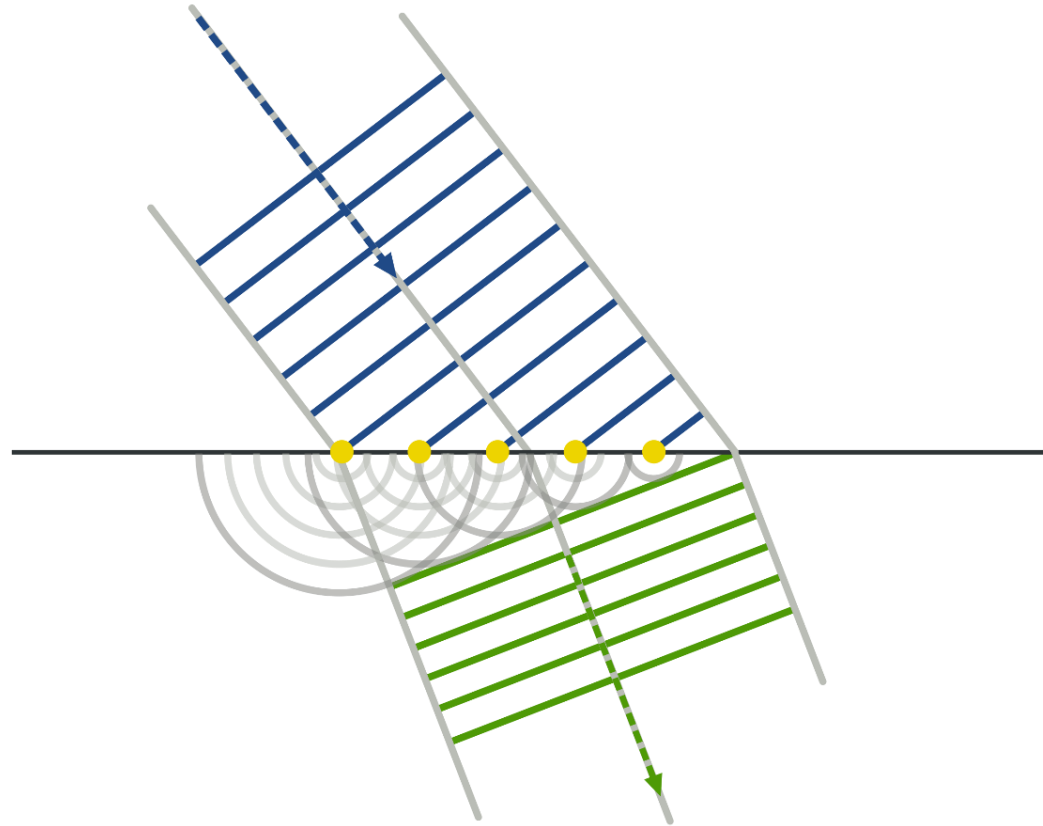
(1704)



Preliminaries: light (wave?)



Refraction



The Huygens–Fresnel principle: every point on a wavefront is itself the source of spherical wavelets, and the secondary wavelets emanating from different points mutually interfere. The sum of these spherical wavelets forms the wavefront.



Robert Hooke???
(1635-1703)



Christiaan Huygens
(1629-1695)

Preliminaries: wave motion



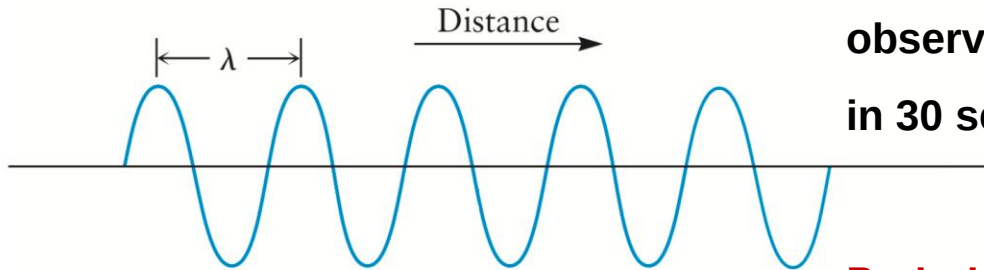
Amplitude (A) 振幅 the maximum height compared with the undisturbed level.

Wavelength (λ) 波长 the distance between two crests.

Frequency (ν) 频率 the number of peaks past a fixed point in space per second.

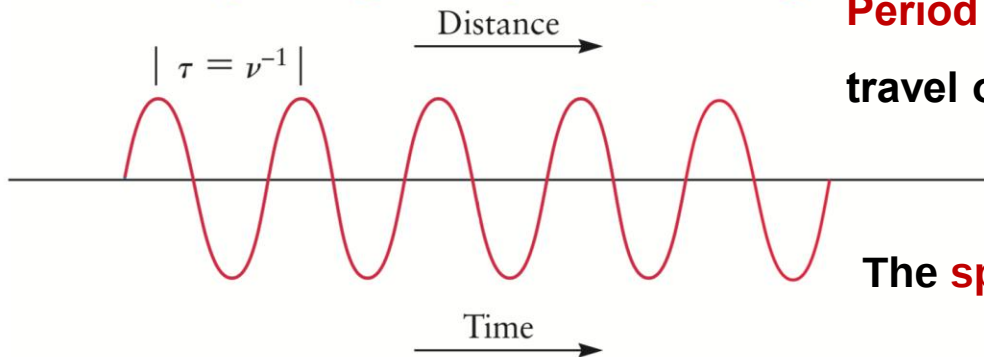
If 12 water-wave peaks are
observed to pass a certain point
in 30 seconds, the frequency

$$\text{frequency} = \nu = \frac{12}{30 \text{ s}} = 0.40 \text{ s}^{-1} = 0.40 \text{ Hz}$$



Period 周期 time required to
travel one wavelength distance

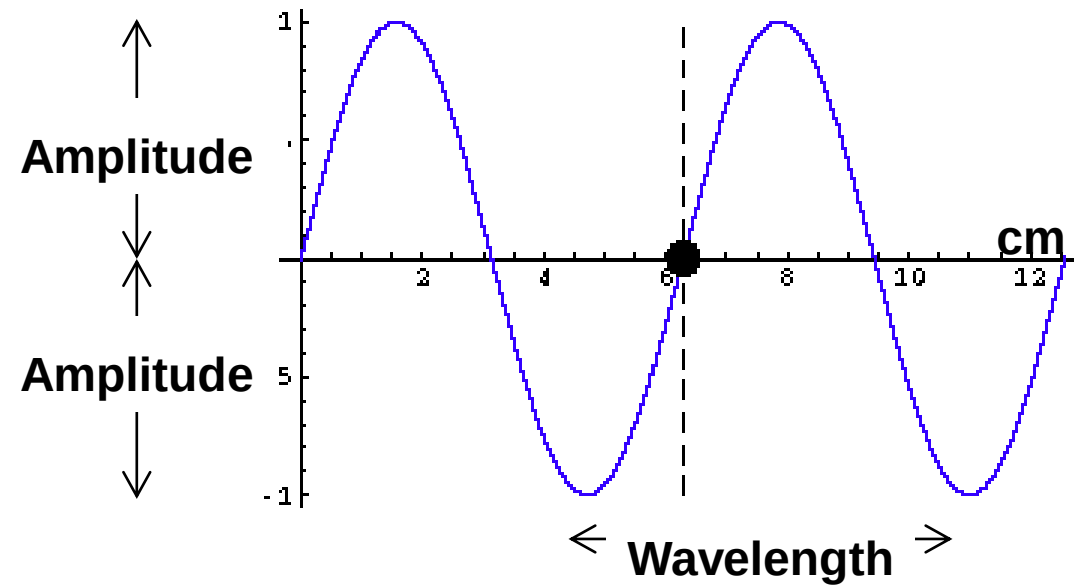
$$\tau = \nu^{-1}$$



The **speed 速度** of a wave

$$\text{speed} = \frac{\text{distance traveled}}{\text{time elapsed}} = \frac{\lambda}{\nu^{-1}} = \lambda \nu$$

Preliminaries: wave motion



Amplitude 振幅

$$A = 1$$

Wavelength 波长

$$\lambda = 6.3 \text{ cm} \leftrightarrow \text{Wavenumber 波数 } \tilde{\nu} = \lambda^{-1} \approx 0.16 \text{ cm}^{-1}$$

Period 周期

$$T = 3.0 \text{ s} \leftrightarrow \text{Frequency 频率 } \nu = T^{-1} \approx 0.33 \text{ Hz}$$

Speed 波速

$$s = \lambda / T = \lambda \nu \approx 2.1 \text{ cm} \cdot \text{s}^{-1}$$

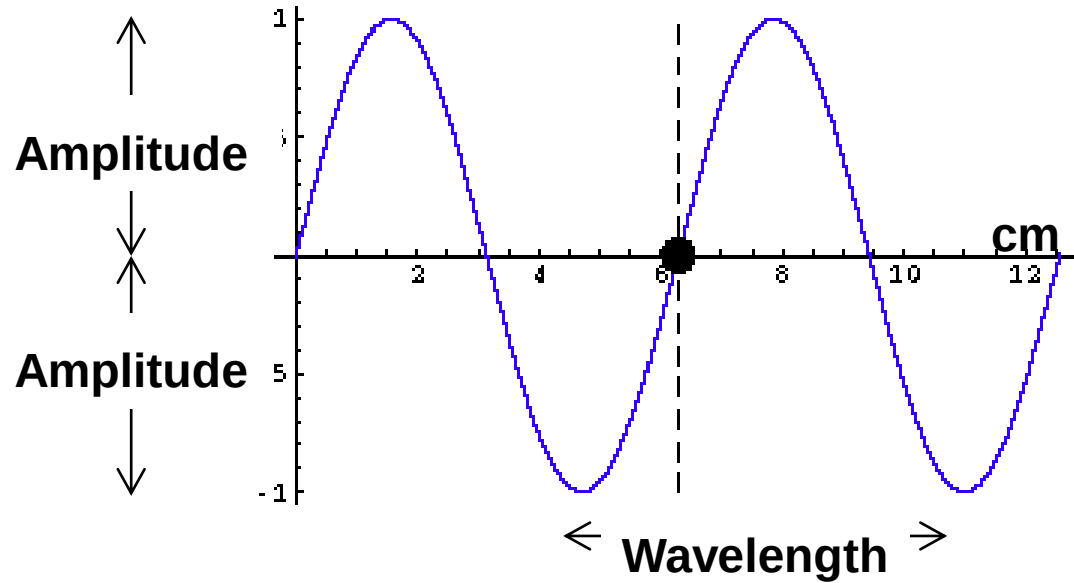
Preliminaries: wave motion

sine wave 正弦波

Phase (φ) propagates in both time and space.

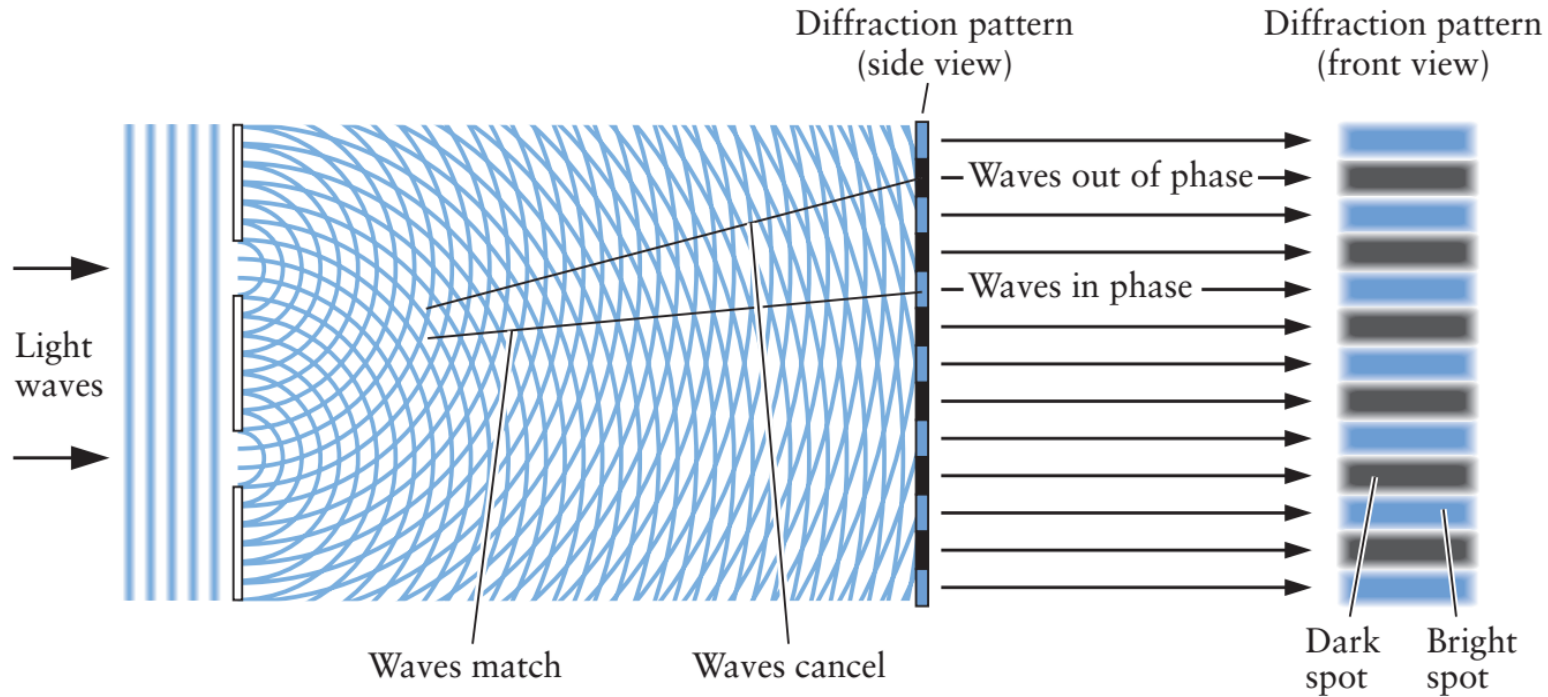
$$\varphi(x) = \frac{2\pi x}{\lambda} = 2\pi \tilde{\nu} x$$

$$\varphi(t) = -\frac{2\pi t}{T} = -2\pi \nu t$$



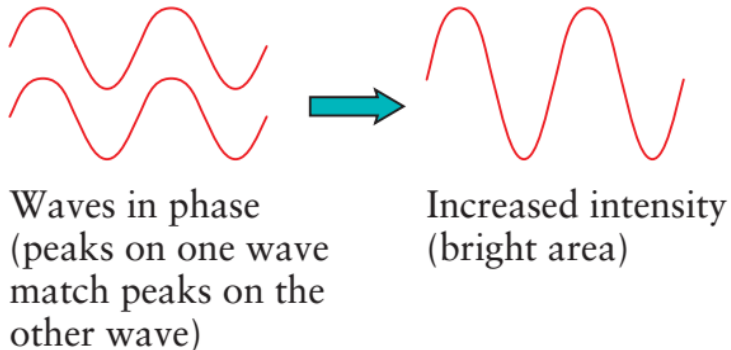
$$y(x,t) = A \sin [\varphi(x) + \varphi(t)] = A \sin \left[2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) \right].$$

Preliminaries: light (wave?)

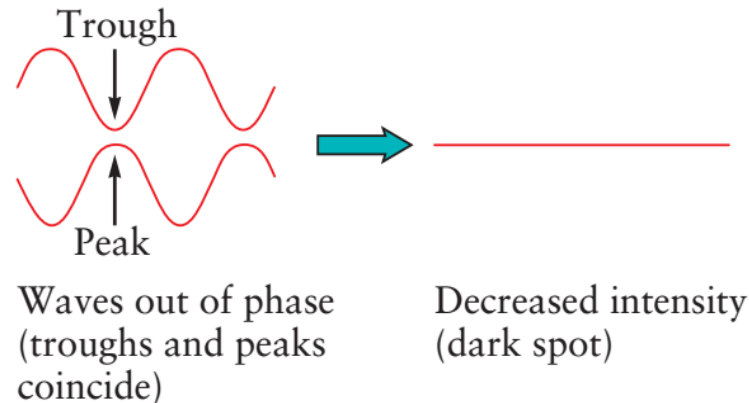


Double-slit experiment, Wave interference

(a) Constructive interference



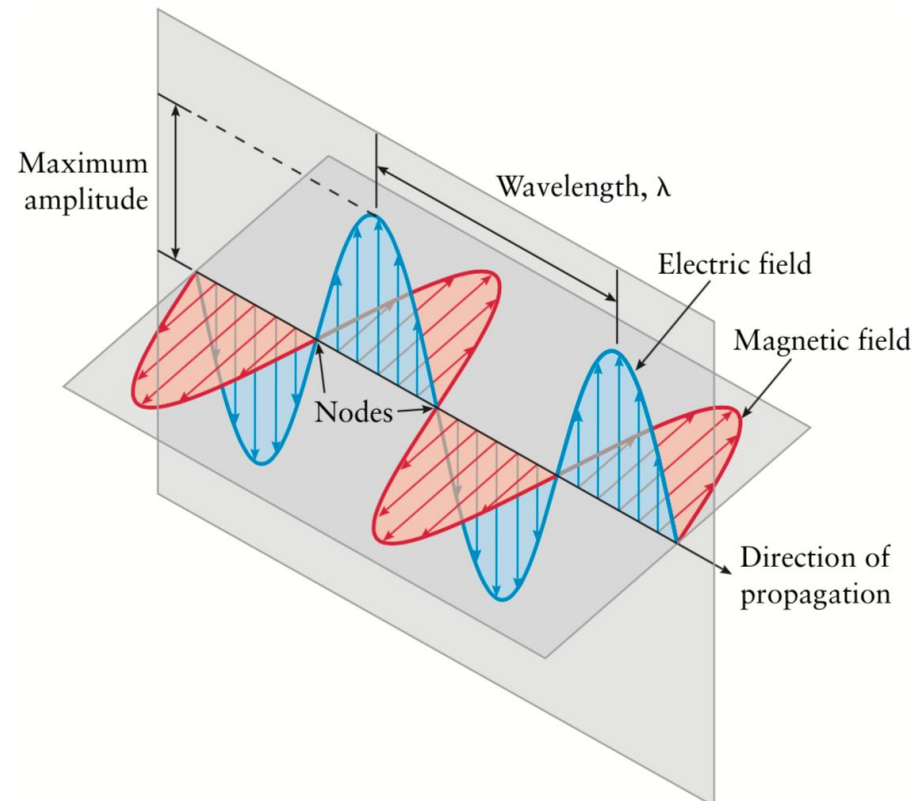
(b) Destructive interference



Thomas Young
(1773-1829)

Preliminaries: light (electromagnetic radiation?)

Name	Integral equations	Differential equations
Gauss's law	$\oiint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \iiint_{\Omega} \rho dV$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$
Gauss's law for magnetism	$\oiint_{\partial\Omega} \mathbf{B} \cdot d\mathbf{S} = 0$	$\nabla \cdot \mathbf{B} = 0$
Maxwell–Faraday equation (Faraday's law of induction)	$\oint_{\partial\Sigma} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
Ampère's circuital law (with Maxwell's addition)	$\oint_{\partial\Sigma} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(\iint_{\Sigma} \mathbf{J} \cdot d\mathbf{S} + \epsilon_0 \frac{d}{dt} \iint_{\Sigma} \mathbf{E} \cdot d\mathbf{S} \right)$	$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$



- Predicted the existence of electromagnetic radiation.
- Predicted visible light is a form of electromagnetic radiation.

Visible light as a propagating wave of **electromagnetic** carries both **energy** and **momentum**.



James Clerk Maxwell
(1831-1879)

Almost all commercially available microwave ovens use radiation with a frequency of 2.45×10^9 Hz. Calculate the wavelength of this radiation.

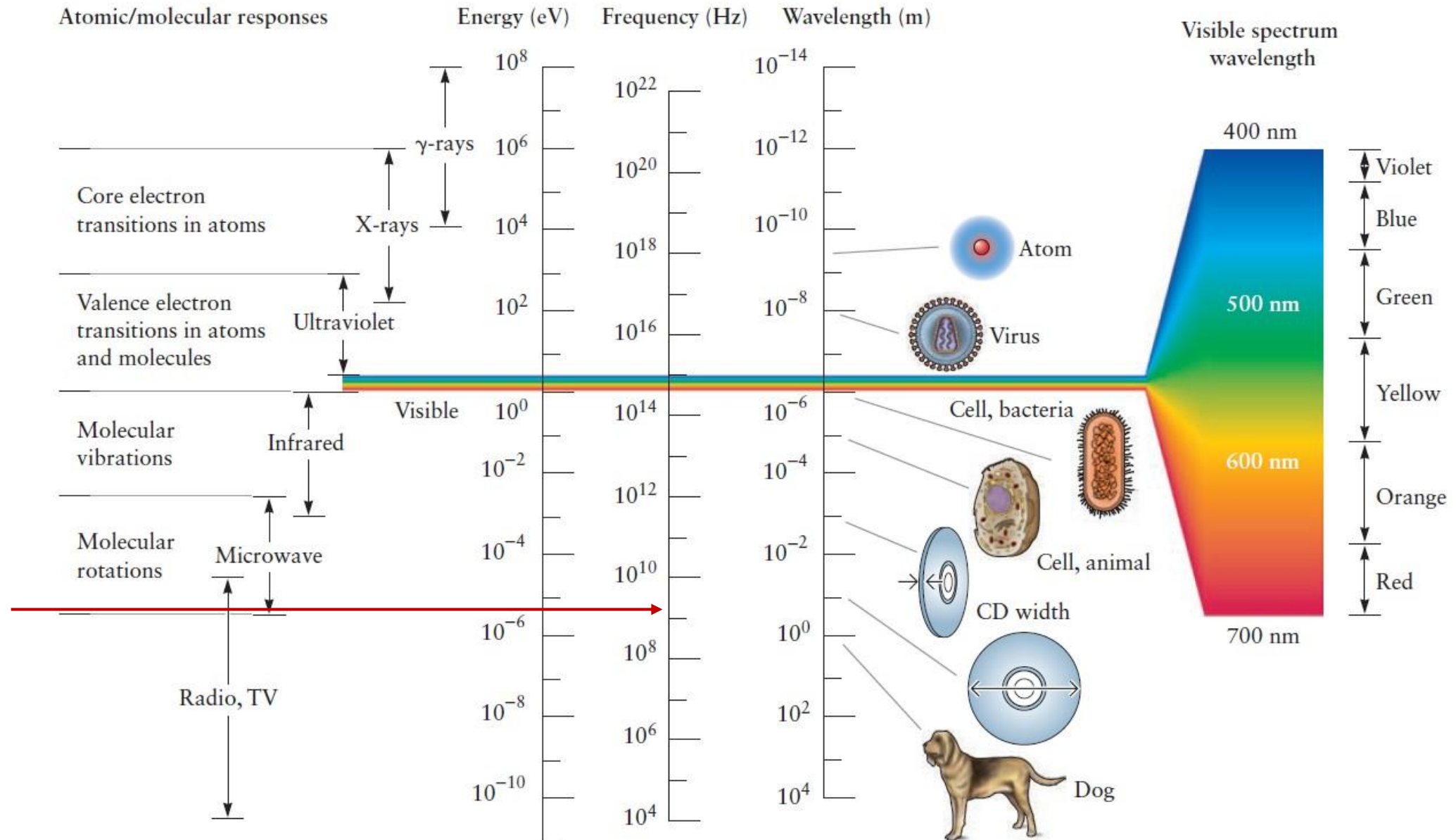
Solution

The wavelength is related to the frequency as follows:

$$\lambda = \frac{c}{\nu} = \frac{3.00 \times 10^8 \text{ s}^{-1}}{2.45 \times 10^9 \text{ s}^{-1}} = 0.122 \text{ m}$$

Thus, the wavelength is 12.2 cm.

Preliminaries: light (electromagnetic radiation?)



Preliminaries: light (electromagnetic radiation?)

The speed, c , of light passing through a vacuum is equal to the product $\lambda\nu$, and its value by **definition** is

$$c = \lambda\nu = 2.99792458 \times 10^8 \text{ m s}^{-1}$$



A laser emits nearly monochromatic light



light contains the full range of visible wavelengths;
it can be resolved into its component wavelengths
by passing it through a prism

THE SI

The SI — the modern metric system — has seven base units from which all other measurement units can be derived. On May 20, 2019, four of them — the kilogram, kelvin, ampere and mole — were redefined in terms of constants of nature. The remaining three — the second, meter, and candela — are already based on universal constants.

Click on the SI symbols below for more information.

Kilogram
kg

Candela
cd

Kelvin
K

Meter
m

Second
s

Ampere
A

Mole
mol

Length: how long is one meter?

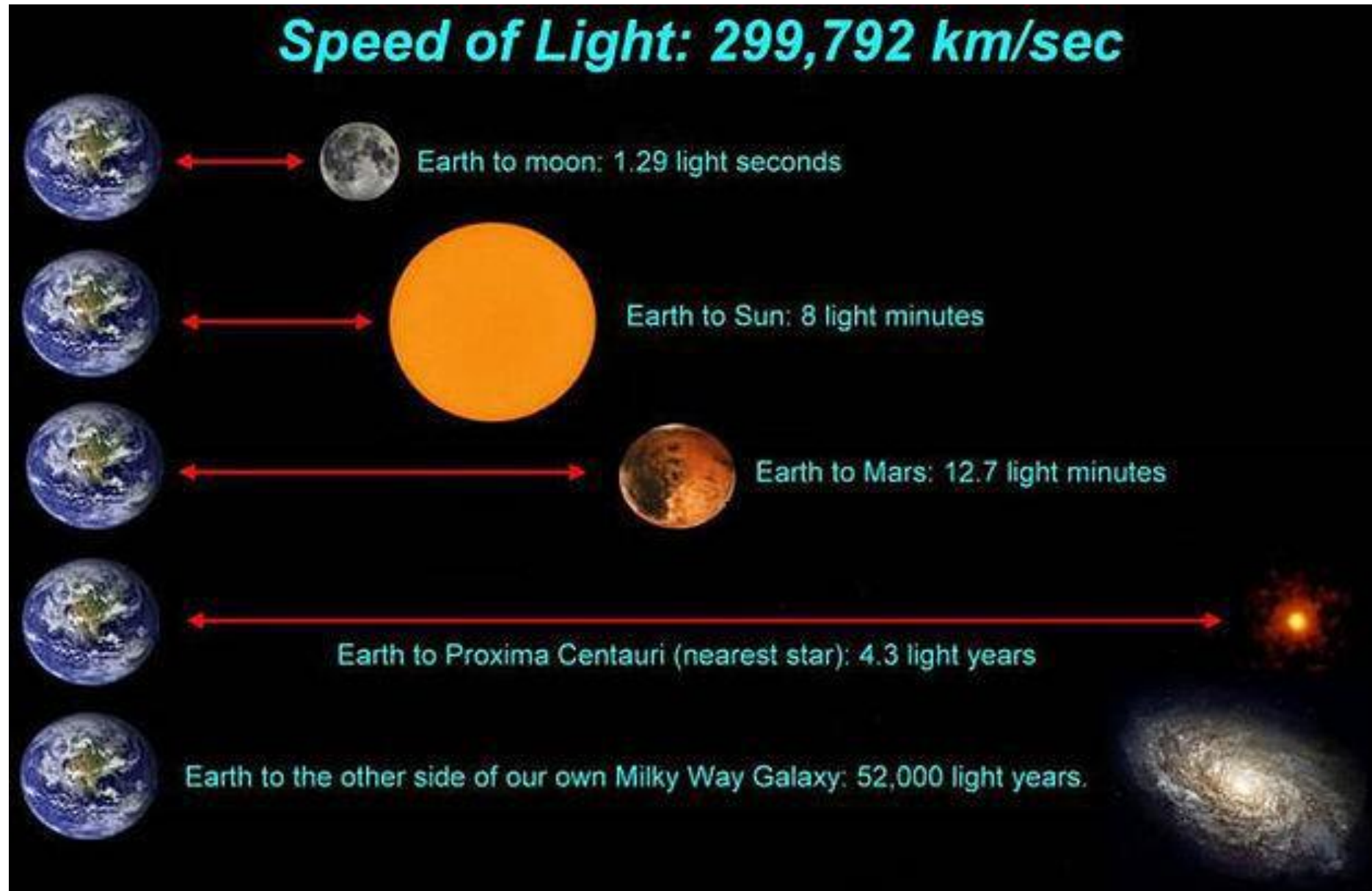
- **Y1791: one ten-millionth of the length of a great circle quadrant along the Earth's meridian through Paris**
- **Y1799: platinum meter bar**
- **Y1983: the length of the path traveled by light in a vacuum in $1/299,792,458$ of a second**



“坐地日行八万里，巡天遥看一千河”



Preliminaries: light (electromagnetic radiation?)



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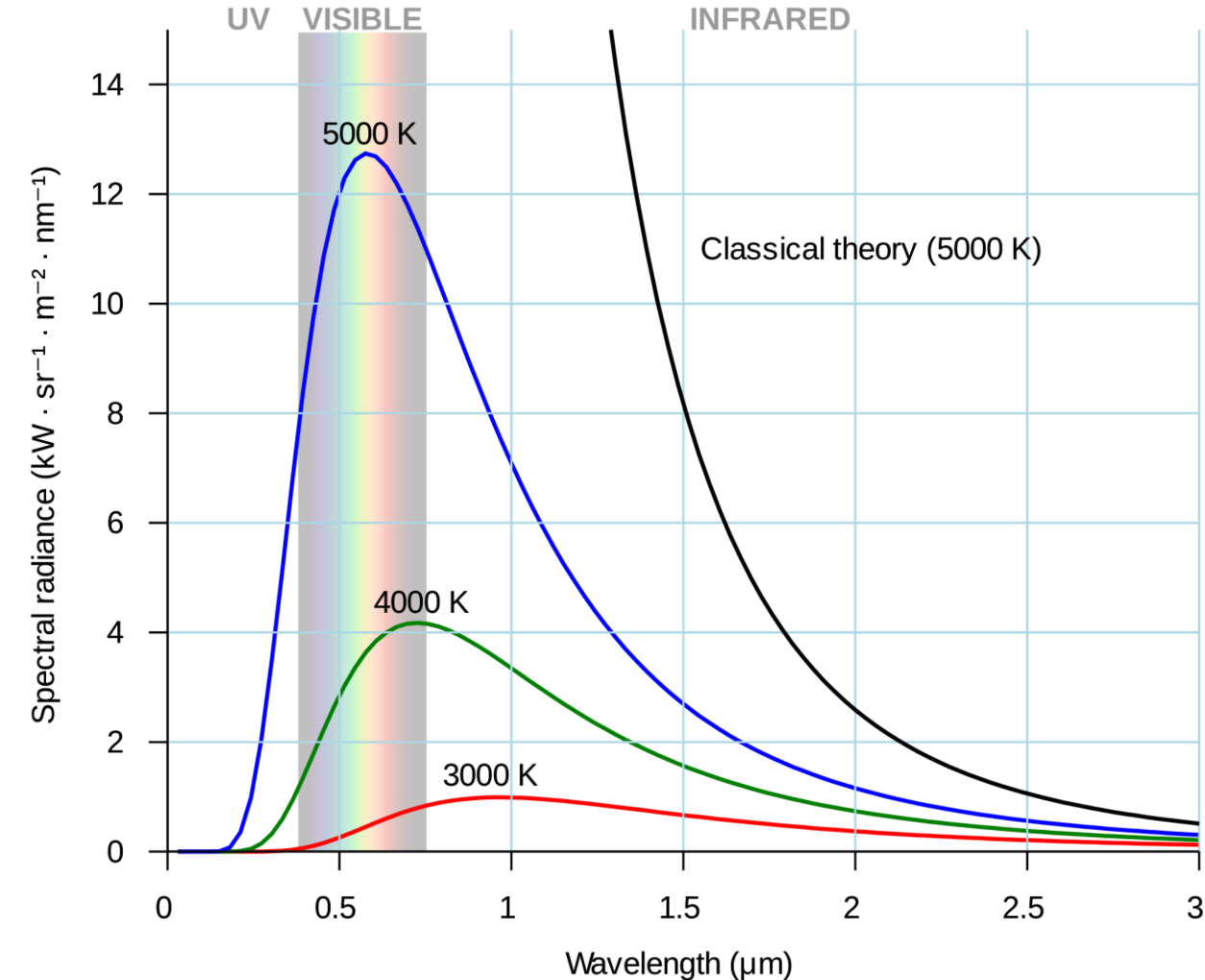
Blackbody Radiation (黑体辐射)



← Warm White Daylight White Cool White →
2700K-3300K 4200-4500K 5500-7000K

- Every object emits energy from its surface in the form of thermal radiation.
- The energy is carried by electromagnetic waves.
- The distribution of the wavelengths depends on the temperature of the object.

Blackbody Radiation



Blackbody: an idealized physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence.

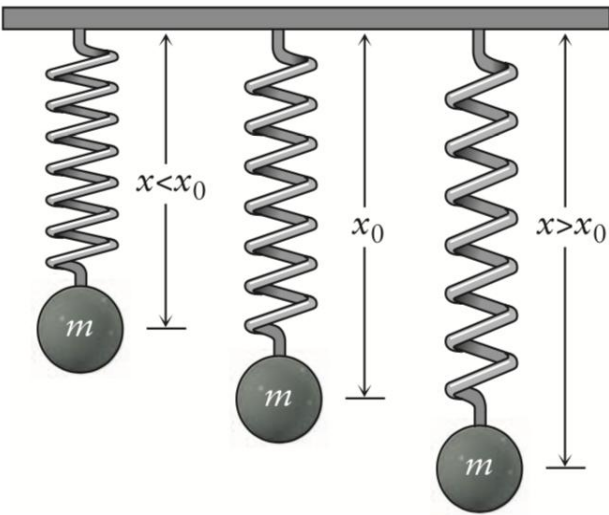
A blackbody is not necessary black in color.

A black body in thermal equilibrium (that is, at a constant temperature) emits electromagnetic radiation called **blackbody radiation**. It has a spectrum that is determined by the temperature alone, not by the body's shape or composition.

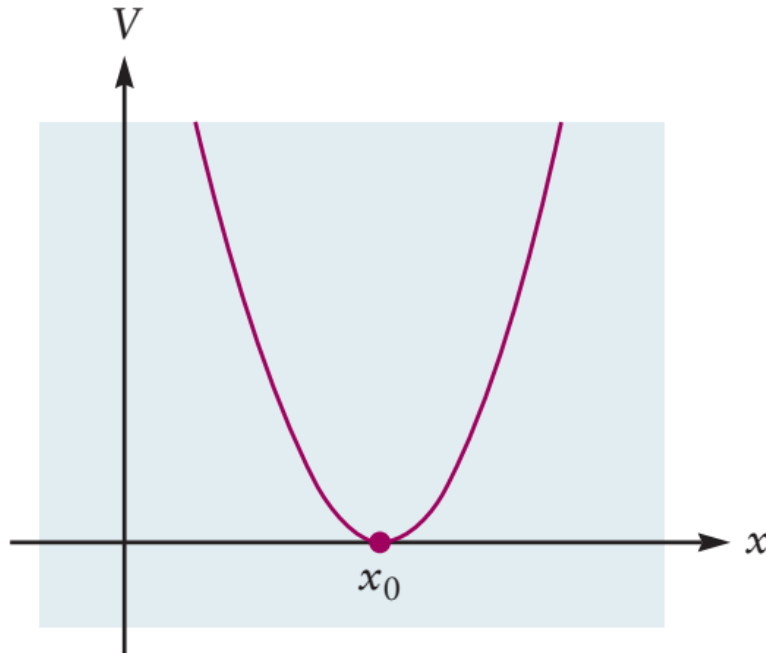
Blackbody Radiation

Classical physics:

Sources of blackbody radiation are oscillating (振蕩) electrical charges in the surfaces of these objects that have been accelerated by ordinary thermal motion.



A charged particle of mass m bound to a solid surface by a spring is a model for the oscillatory motions of the surface atoms of a black body. The particle is shown at its rest position x_0 , at a position closer to the surface ($x < x_0$), and at one further from the surface ($x > x_0$).



$$F = -k(x - x_0)$$

$$V(x) = \frac{1}{2} k(x - x_0)^2$$

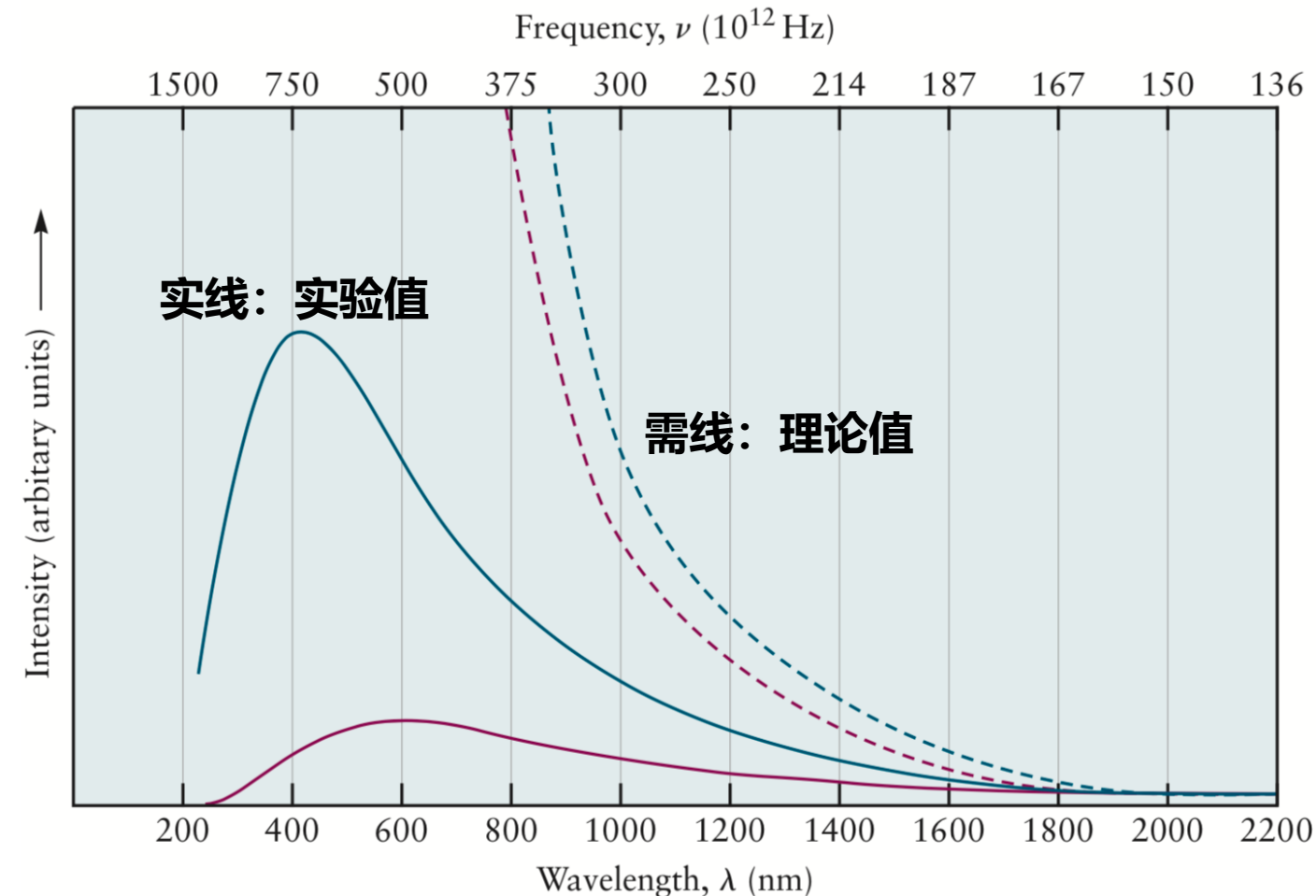
Classical mechanics puts no restrictions on the value of the total energy, E . **The total energy can be large or small, and it can be changed smoothly and continuously from one arbitrary value to another.**

Blackbody Radiation

Classical physics:
$$\rho_T(\nu) = \frac{8\pi k_B T \nu^2}{c^3}$$

Rayleigh-Jeans law

- $\rho_T(\nu)$ is the intensity of the radiation at the frequency ν ;
- k_B is a fundamental constant called the Boltzmann constant
- T is the temperature in kelvins (K)
- c is the speed of light.

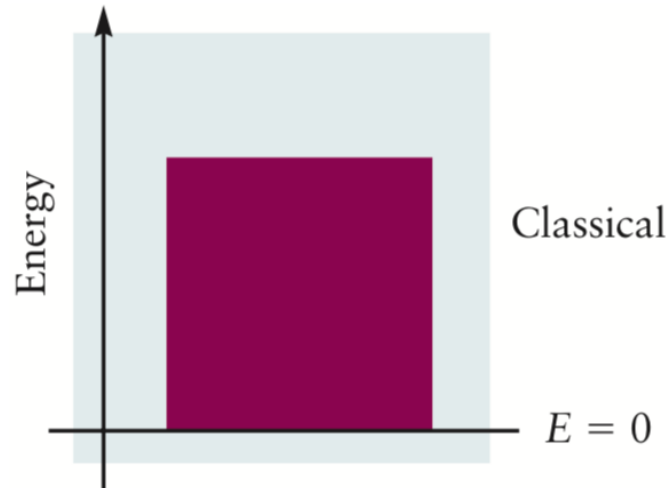


“ultraviolet catastrophe”: predicts an infinite intensity at very short wavelengths, whereas the experimental intensities always remain finite and actually fall to zero at very short wavelengths.

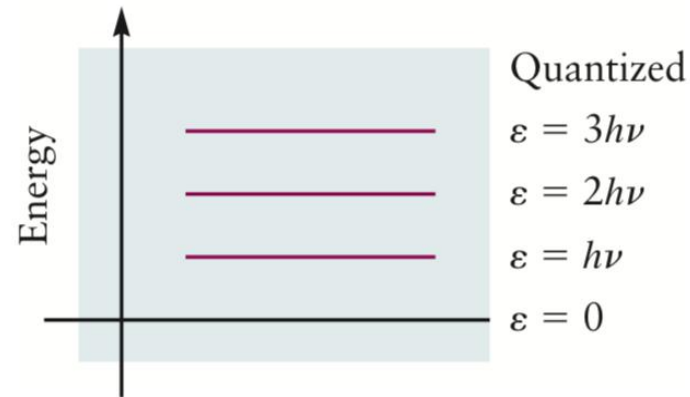
Blackbody Radiation

Planck's 1st hypothesis: It is not possible to put an arbitrary amount of energy into an oscillator of frequency. Instead, **the oscillator must gain and lose energy in “packets,” or quanta, of magnitude $h\nu$.** The total energy of an oscillator can take only discrete values that are integral multiples of $h\nu$.

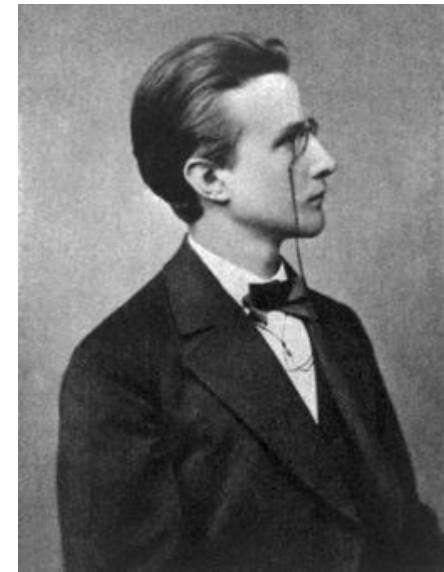
$$\epsilon_{\text{osc}} = nh\nu \quad n = 1, 2, 3, 4, \dots$$



An oscillator obeying classical mechanics has continuous values of energy and can gain or lose energy in arbitrary amounts.



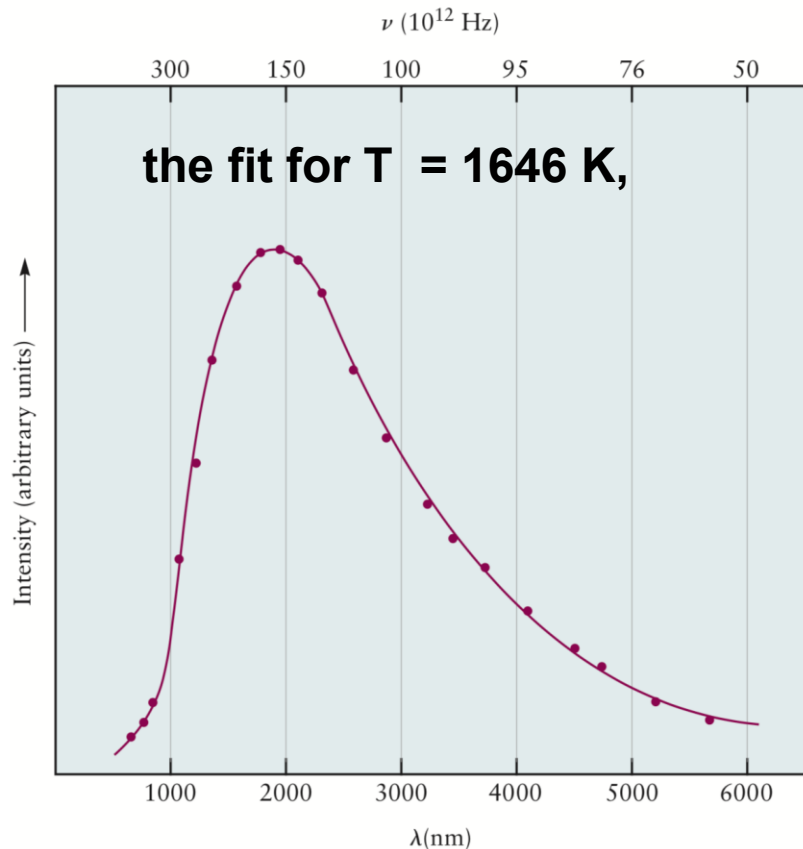
An oscillator described by Planck's postulate has discrete energy levels. It can gain or lose energy only in amounts that correspond to the difference between two energy levels.



**Max Planck
(1858-1947)**

Blackbody Radiation

Planck's 2nd step: the excitation of a particular oscillator is an **all-or-nothing event**. The falloff in intensity with frequency at a given temperature of the blackbody radiation is due to a diminishing probability of exciting the high-frequency oscillators.



$$\rho_T(\nu) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$$



Max Planck
(1858–1947)

Planck's constant

**In the revised SI, the Planck constant h is equal to exactly
 $6.626\,070\,15 \times 10^{-34}$ Joule seconds**



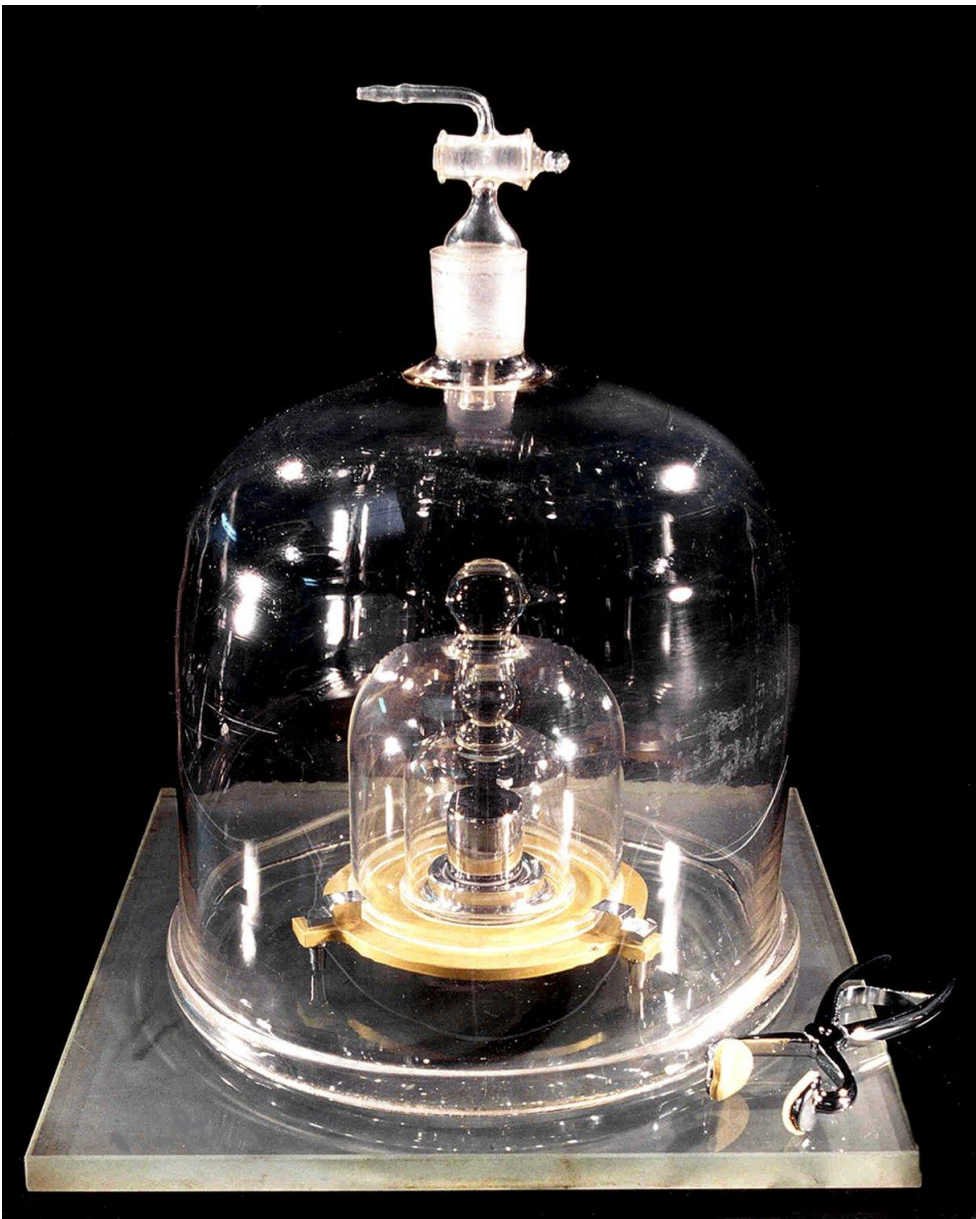
Photo from the Nobel
Foundation archive.

Max Karl Ernst
Ludwig Planck

Prize share: 1/1

The Nobel Prize in Physics 1918 was awarded to
Max Karl Ernst Ludwig Planck “in recognition of
the services he rendered to the advancement of
Physics by his discovery of energy quanta”.

<https://www.nobelprize.org/prizes/physics/1918/summary/>



One Mass to Rule Them All



'One Ring to rule them all, One Ring to find them,
One Ring to bring them all and in the darkness bind them
Three Rings for the Elven-kings under the sky,
Seven for the Dwarf-lords in their halls of stone,
Nine for Mortal Men doomed to die,
One for the Dark Lord on his dark throne
In the Land of Mordor where the Shadows lie.
One Ring to rule them all, One Ring to find them,
One Ring to bring them all and in the darkness bind them
In the Land of Mordor where the Shadows lie."
(J.R.R. Tolkien)

One Ring to Rule Them All

Weight: how heavy is one kg?

Y1793: the mass of 1 dm³ of water

Y1795: at the melting point of ice

Y1889: under atmospheric pressure and at the temperature of its maximum density (and a prototype kg)

Y2019: Planck constant

THE SI

The SI — the modern metric system — has seven base units from which all other measurement units can be derived. On May 20, 2019, four of them — the kilogram, kelvin, ampere and mole — were redefined in terms of constants of nature. The remaining three — the second, meter, and candela — are already based on universal constants.

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Second
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Ampere
A

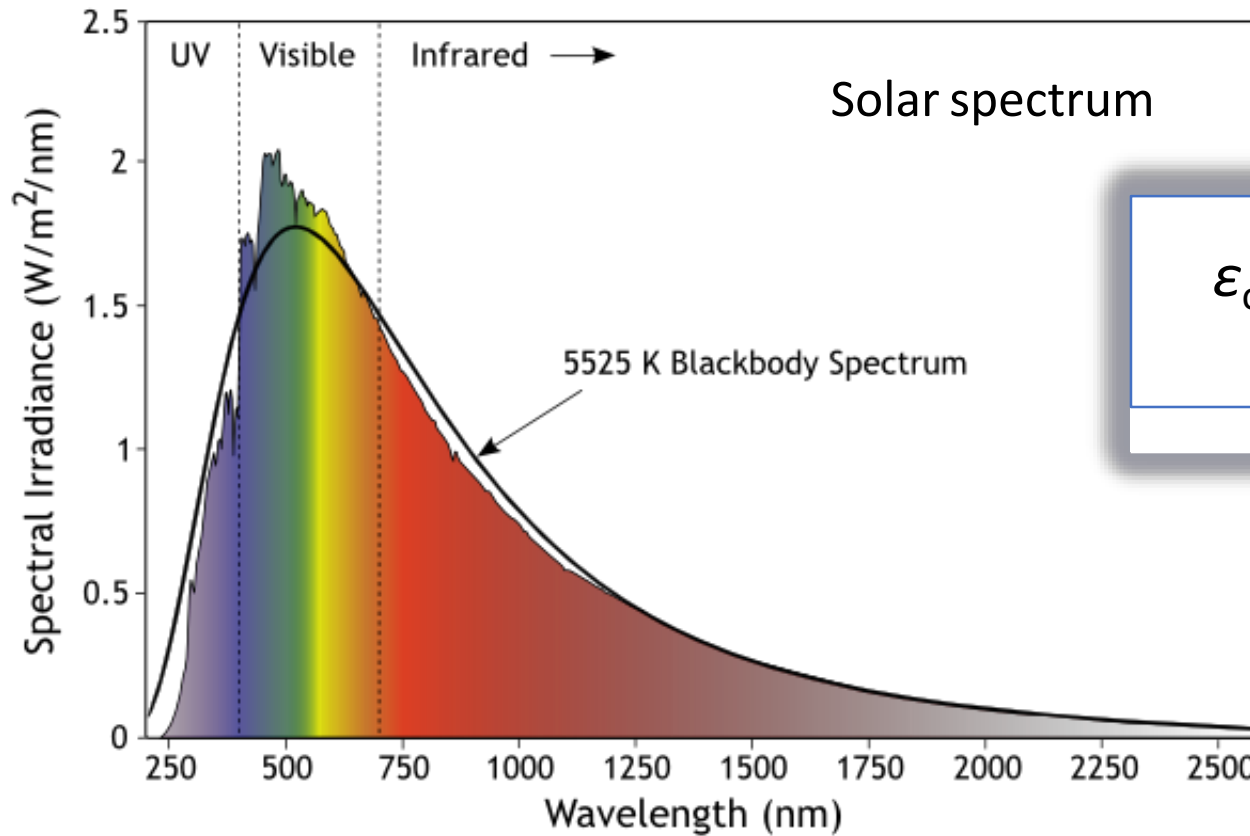
Mole
mol

Blackbody Radiation

Planck's dramatic explanation of blackbody radiation includes **three fundamentally new ideas**:

- The energy of a system can take only discrete values, which are represented on its energy-level diagram.
- A quantized oscillator can gain or lose energy only in discrete amounts $\Delta\varepsilon$, which are related to its frequency by $\Delta\varepsilon = h\nu$.
- To emit energy from higher energy states, the temperature of a quantized system must be sufficiently high to excite those states.

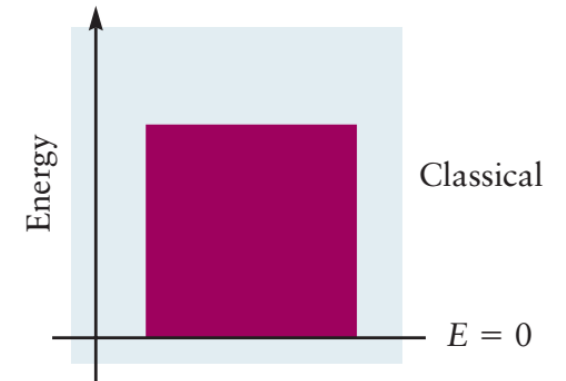
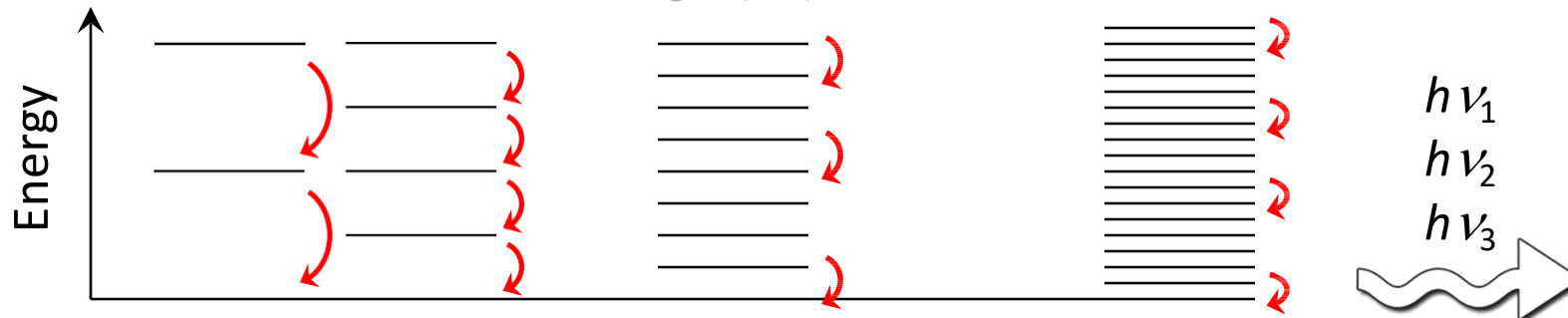
Blackbody Radiation



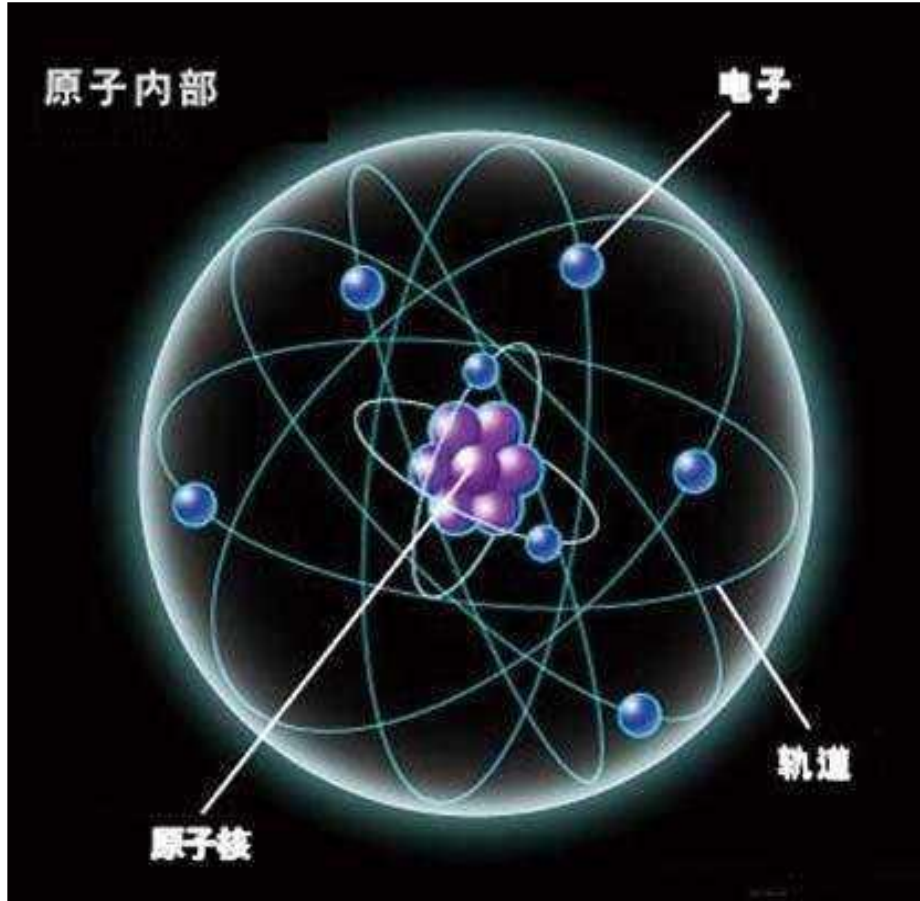
$$\nu = \frac{c}{\lambda}$$

$$\epsilon_{\text{osc}} = E = n h \nu, n \in \mathbb{N}$$

$$h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$$



Evidence for energy quantization in atoms



According to Maxwell's electromagnetic theory:
accelerated charges must emit electromagnetic radiation

An **electron in orbit** around the nucleus is accelerating because its **direction is constantly changing**. It must, therefore, **emit electromagnetic radiation**, lose energy, and eventually spiral into the nucleus.

Rutherford's planetary model

Atomic spectra



Ne



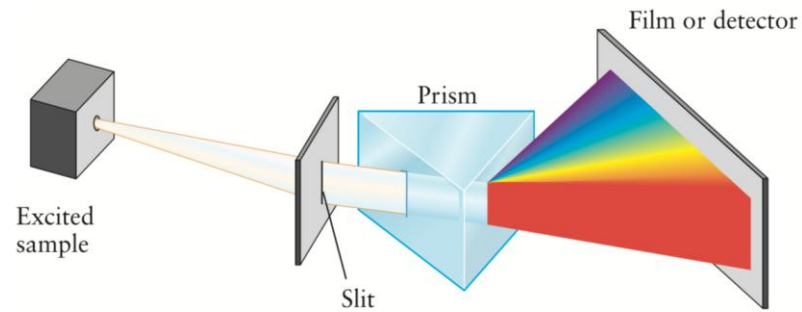
Ar



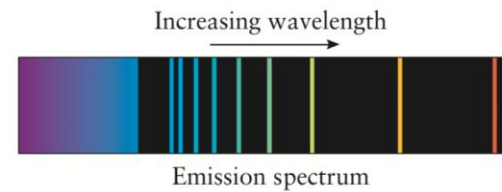
Hg

One instrument used to separate light into its component wavelengths is called a spectrograph

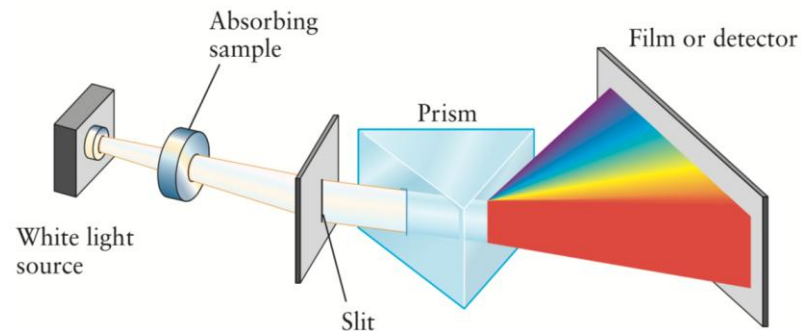
光谱仪



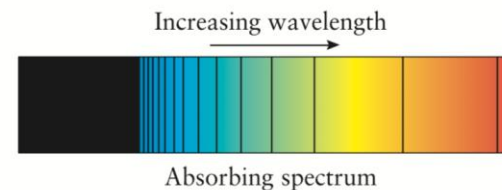
(a)



Emission spectrum



spectrum, 光谱



Absorption spectrum

Atomic spectra

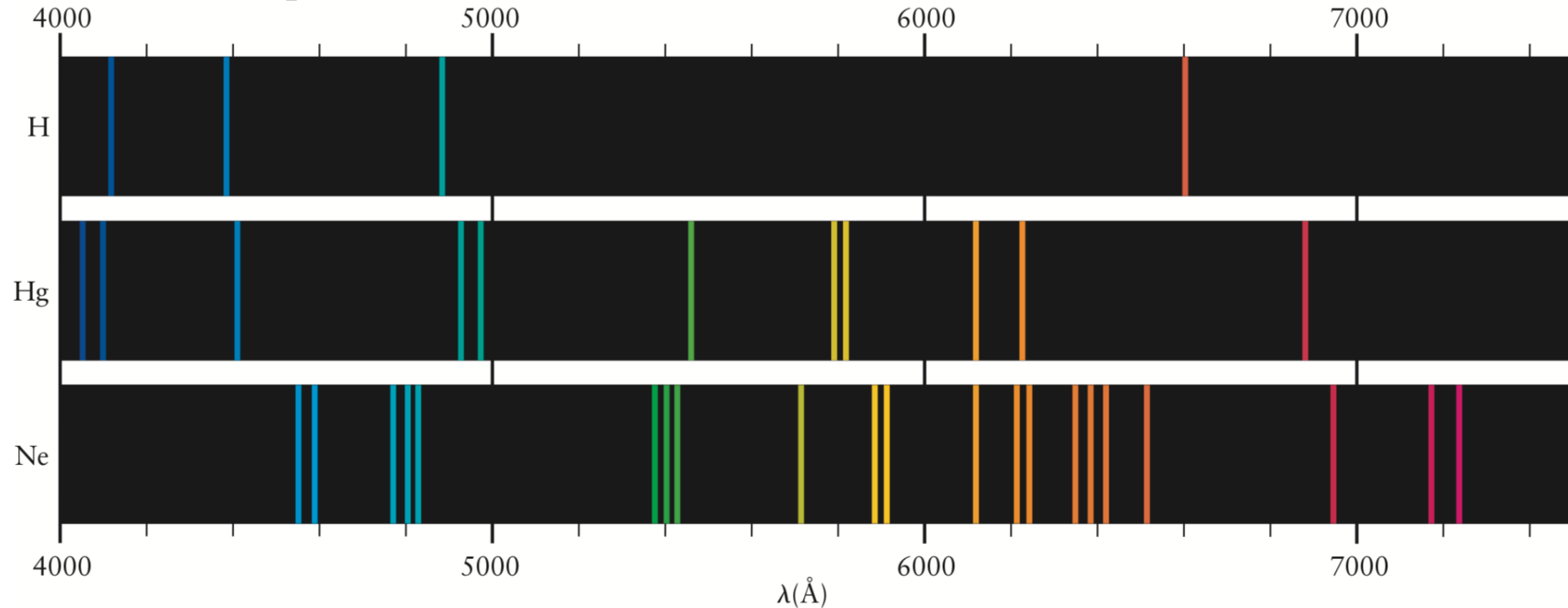
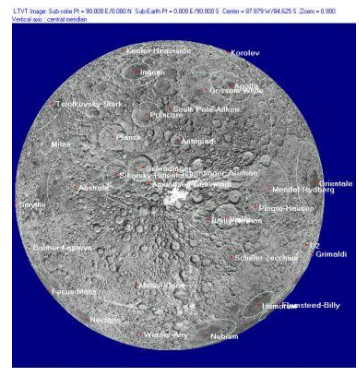


FIGURE 4.10 Atoms of hydrogen, mercury, and neon emit light at discrete wavelengths. The pattern seen is characteristic of the element under study. $1 \text{ \AA} = 10^{-10} \text{ m}$.

1885, J. J. Balmer discovered that hydrogen atoms emit a series of lines in the visible region, with frequencies given by the following simple formula:

$$\nu = \left[\frac{1}{4} - \frac{1}{n^2} \right] \times 3.29 \times 10^{15} \text{ s}^{-1} \quad n = 3, 4, 5, \dots$$



Atomic spectra



$$\frac{656}{365} \approx 1.80 = \frac{9}{5} = \frac{9}{9-4}$$

$$\frac{486}{365} \approx 1.33 = \frac{4}{3} = \frac{16}{16-4}$$

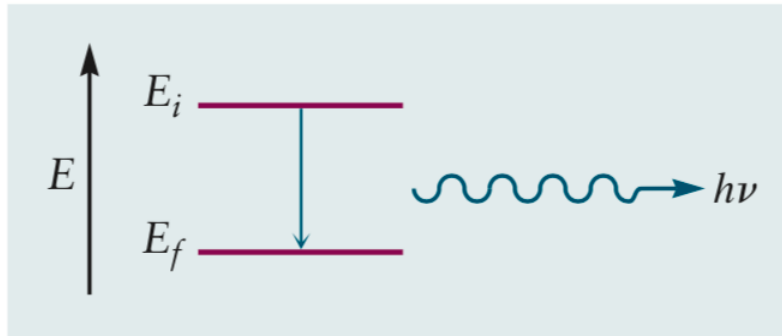
$$\Rightarrow \lambda_n = B \frac{n^2}{n^2-4}, \quad B = 365 \text{ nm}, \quad n = 3, 4, 5, \dots$$

$$\nu = \left[\frac{1}{4} - \frac{1}{n^2} \right] \times 3.29 \times 10^{15} \text{ s}^{-1} \quad n = 3, 4, 5, \dots$$

Atomic spectra

Niels Bohr proposed a model:

1. the hydrogen atom allowed only discrete energy states to exist;
2. light absorption resulted from a transition of the atoms between two of these states.



$$\nu = \frac{E_f - E_i}{h} \quad \text{or} \quad \Delta E = h\nu$$

h is Planck's constant



Niels Bohr
(1885-1962)

- In an atom, there are various states 能态 of fixed, discrete energy.
- Emission or absorption of light results from a transition 跃迁 between two of these states.
- The energy quantum $h\nu$ of the absorbed/emitted light equals the transition energy.



Photo from the Nobel
Foundation archive.

Niels Henrik David
Bohr

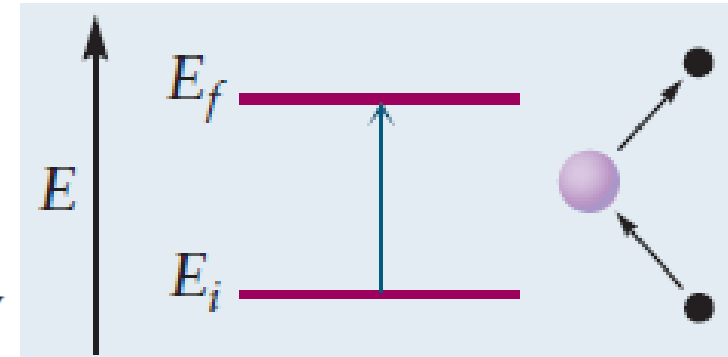
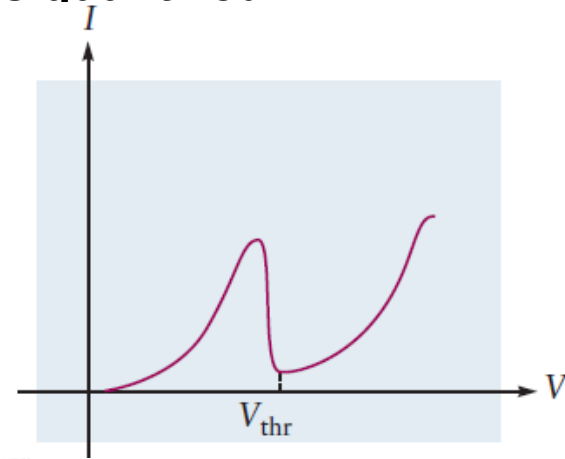
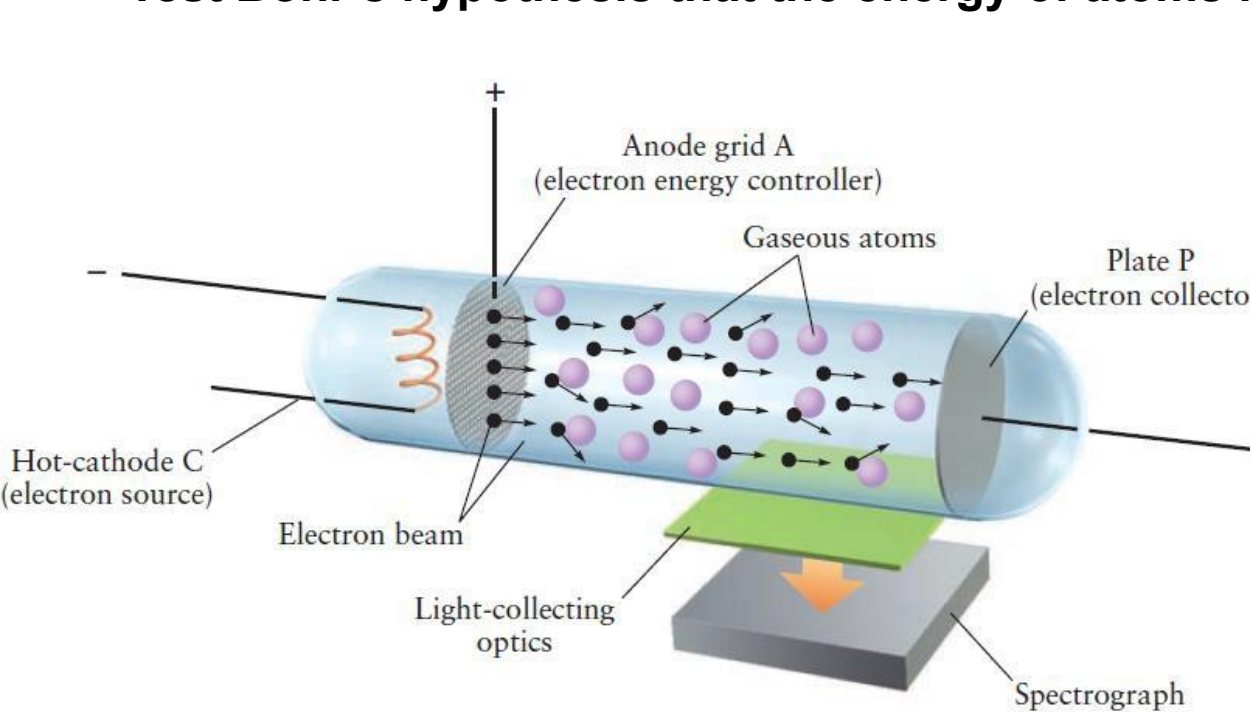
Prize share: 1/1

The Nobel Prize in Physics 1922 was awarded to
Niels Henrik David Bohr "for his services in the
investigation of the structure of atoms and of the
radiation emanating from them"

<https://www.nobelprize.org/prizes/physics/1922/summary/>

The Franck–Hertz Experiment

Test Bohr's hypothesis that the energy of atoms is quantized



The current shows a sharp change at a particular value of the accelerating voltage, corresponding to the threshold energy transfer from the electron to a gaseous atom.

When the accelerating voltage was slightly above V_{thr} , a single emission line was observed whose frequency was very nearly equal to

$$\nu = \frac{\Delta E}{h} = \frac{eV_{thr}}{h}$$

The first two excitation voltage thresholds in the Franck–Hertz study of mercury vapor were found at 4.9 and 6.7 V. Calculate the wavelength of light emitted by mercury atoms after excitation past each of these thresholds.

Solution

The emitted wavelength at a particular value of V_{thr} is given by the following equation:

$$\begin{aligned}\lambda &= \frac{hc}{\Delta E} = \frac{hc}{eV_{\text{thr}}} = \frac{(6.6261 \times 10^{-34} \text{ J s})(2.9979 \times 10^8 \text{ ms}^{-1})}{(1.6022 \times 10^{-19} \text{ C})(V_{\text{thr}}[\text{V}])} \\ &= \frac{1239.8 \text{ nm}}{V_{\text{thr}}[\text{V}]}\end{aligned}$$

The value of each emission wavelength is calculated by substituting in a particular value of V_{thr} expressed in units of volts (V).

At $V_{\text{thr}} = 4.9 \text{ V}$, the calculated wavelength is $\lambda = 250 \text{ nm}$. The wavelength actually observed above this threshold was 253.7 nm.

At $V_{\text{thr}} = 6.7 \text{ V}$, the calculated wavelength is $\lambda = 180 \text{ nm}$. The wavelength actually observed above this threshold was 184.9 nm.

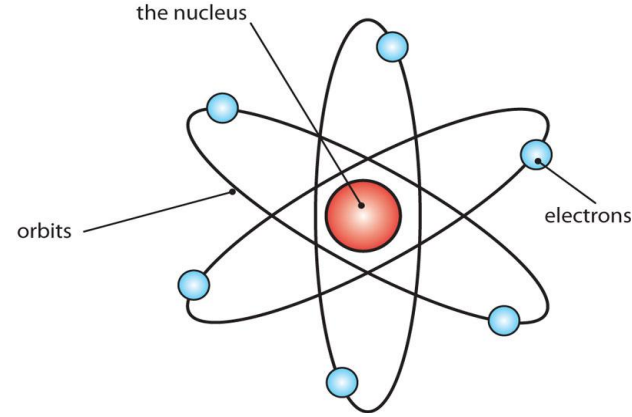
Energy differences measured by the Franck–Hertz method and by optical emission spectroscopy agree quite closely. The optical measurements are more precise. These results enable us to begin to construct the energy diagram for mercury, showing the location of the first two excited states relative to the ground state.

Major points of today's lecture

- **Introduction to quantum mechanics**

1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. **The Bohr model: predicting discrete energy levels in atoms**
4. Evidence for wave-particle duality
5. The Schrödinger equation
6. Quantum mechanics of particle-in-a-box models
7. Wave functions for particles in two- and three-dimensional boxes

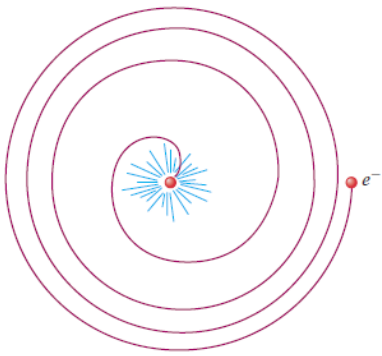
The Bohr model: predicting discrete energy levels in atoms



**Rutherford Model
(1911)**

The total energy of the hydrogen atom:

$$E = \frac{1}{2} m_e v^2 - \frac{Z e^2}{4 \pi \epsilon_0 r}$$



Classical theory states that atoms constructed according to Rutherford's nuclear model are not stable. The motion of electrons around the nucleus would cause them to radiate energy and quickly spiral into the nucleus.

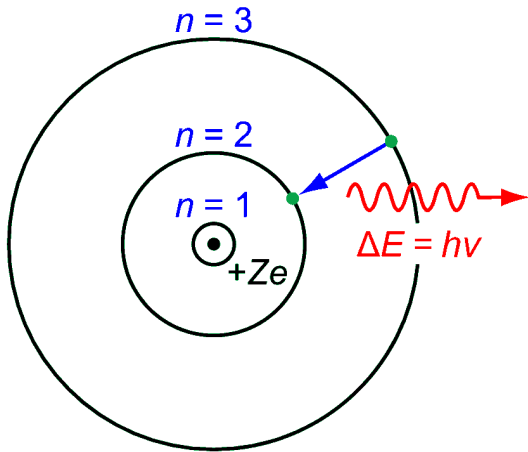
$$|F_{\text{Coulomb}}| = |m_e a|$$



$$\frac{Z e^2}{4 \pi \epsilon_0 r^2} = m_e \frac{v^2}{r}$$

The Bohr model: predicting discrete energy levels in atoms

Bohr Model



Bohr model:

- **only certain discrete orbits are allowed**
(characterized by radius r_n and energy E_n)
- light is emitted or absorbed only when the electron “jumps” from one stable orbit to another

Bohr postulate: **the angular momentum is quantized in integral multiples of $h/2\pi$**

- Linear momentum: $p = m_e v$
- Angular momentum: $L = m_e v r$

$$L = m_e v r = n \frac{h}{2\pi} \quad n = 1, 2, 3, \dots$$

The Bohr model: predicting discrete energy levels in atoms

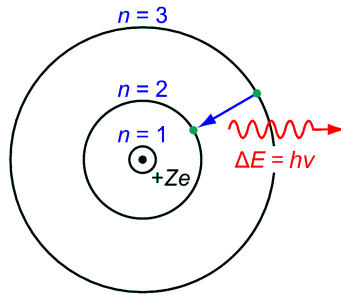
$$L = m_e v r = n \frac{h}{2\pi} \quad n = 1, 2, 3, \dots$$

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m_e \frac{v^2}{r} \quad \longrightarrow \quad r_n = \frac{\epsilon_0 n^2 h^2}{\pi Z e^2 m_e} = \frac{n^2}{Z} a_0$$

a_0 , the **Bohr radius**, has the numerical value $5.29 \times 10^{-11} \text{ m} = 0.529 \text{ \AA}$. $(a_0 = \epsilon_0 h^2 / \pi e^2 m_e)$

The Bohr model: predicting discrete energy levels in atoms

1st prediction: the existence of a series of orbits whose distances from the nucleus increase dramatically with increasing n .



The velocity v_n corresponding to the orbit with radius r_n .

$$r_n = \frac{\epsilon_0 n^2 h^2}{\pi Z e^2 m_e} = \frac{n^2}{Z} a_0$$

$$L = m_e v r = n \frac{h}{2\pi} \quad n = 1, 2, 3, \dots$$

$$v_n = \frac{nh}{2\pi m_e r_n} = \frac{Ze^2}{2\epsilon_0 nh}$$

$$E = \frac{1}{2} m_e v^2 - \frac{Ze^2}{4\pi\epsilon_0 r}$$

2nd prediction: a discrete energy-level diagram for the one-electron atom. The ground state (基态) is identified by $n=1$, and the excited states (激发态) have higher values of n .

$$E_n = \frac{-Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} = -(2.18 \times 10^{-18} \text{ J}) \frac{Z^2}{n^2} \quad n = 1, 2, 3, \dots$$

The Bohr model: predicting discrete energy levels in atoms

Allow values of the energy

$$E_n = \frac{-Z^2 e^4 m_e}{8 \epsilon_0^2 n^2 h^2} = -(2.18 \times 10^{-18} \text{ J}) \frac{Z^2}{n^2} \quad n = 1, 2, 3, \dots$$

$$E_n = -\frac{Z^2}{n^2} \quad (\text{rydberg}) \quad n = 1, 2, 3, \dots$$

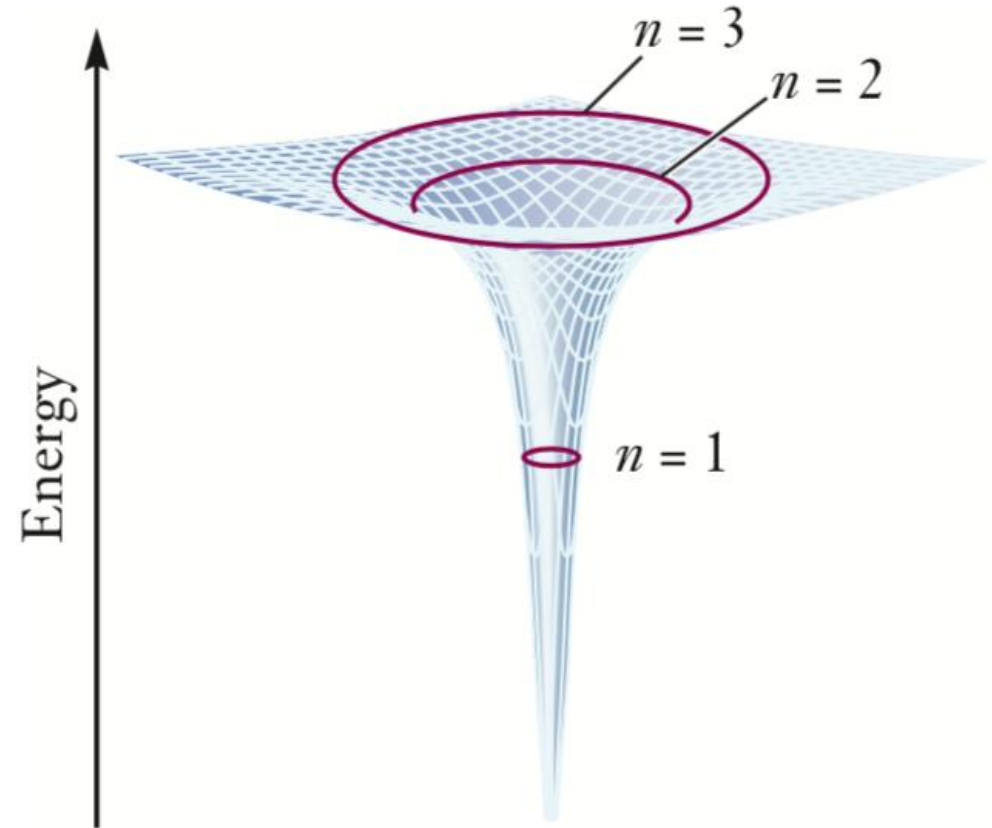
$$1 \text{ rydberg} = 2.18 \times 10^{-18} \text{ J}$$

The ionization energy:

$$\Delta E = E_{\text{final}} - E_{\text{initial}} = 0 - (-2.18 \times 10^{-18} \text{ J}) = 2.18 \times 10^{-18} \text{ J}$$

$$IE = (6.022 \times 10^{23} \text{ atoms mol}^{-1})(2.18 \times 10^{-18} \text{ J atom}^{-1}) = 1.31 \times 10^6 \text{ J mol}^{-1} = 1310 \text{ kJ mol}^{-1}$$

This prediction agrees with the experimentally observed ionization energy of hydrogen atoms and provides confidence in the validity of the Bohr model.



Consider the $n = 2$ state of Li^{2+} . Using the Bohr model, calculate the radius of the electron orbit, the electron velocity, and the energy of the ion relative to that of the nucleus and electron separated by an infinite distance.

Solution

Because $Z = 3$ for Li^{2+} (the nuclear charge is $+3e$) and $n = 2$, the radius is

$$r_2 = \frac{n^2}{Z} a_0 = \frac{4}{3} a_0 = \frac{4}{3} (0.529 \text{ \AA}) = 0.705 \text{ \AA}$$

The velocity is

$$v_2 = \frac{nh}{2\pi m_e r_2} = \frac{2(6.626 \times 10^{-34} \text{ J s})}{2\pi(9.11 \times 10^{-31} \text{ kg})(0.705 \times 10^{-10} \text{ m})} = 3.28 \times 10^6 \text{ m s}^{-1}$$

The energy is

$$E_2 = -\frac{(3)^2}{(2)^2} (2.18 \times 10^{-18} \text{ J}) = -4.90 \times 10^{-18} \text{ J}$$

Typically, atomic sizes fall in the range of angstroms, and atomic excitation energies in the range of 10^{-18} J. This is consistent with the calculations of coulomb potential energies in electron volts and dimensions in angstroms in Section 3.3.

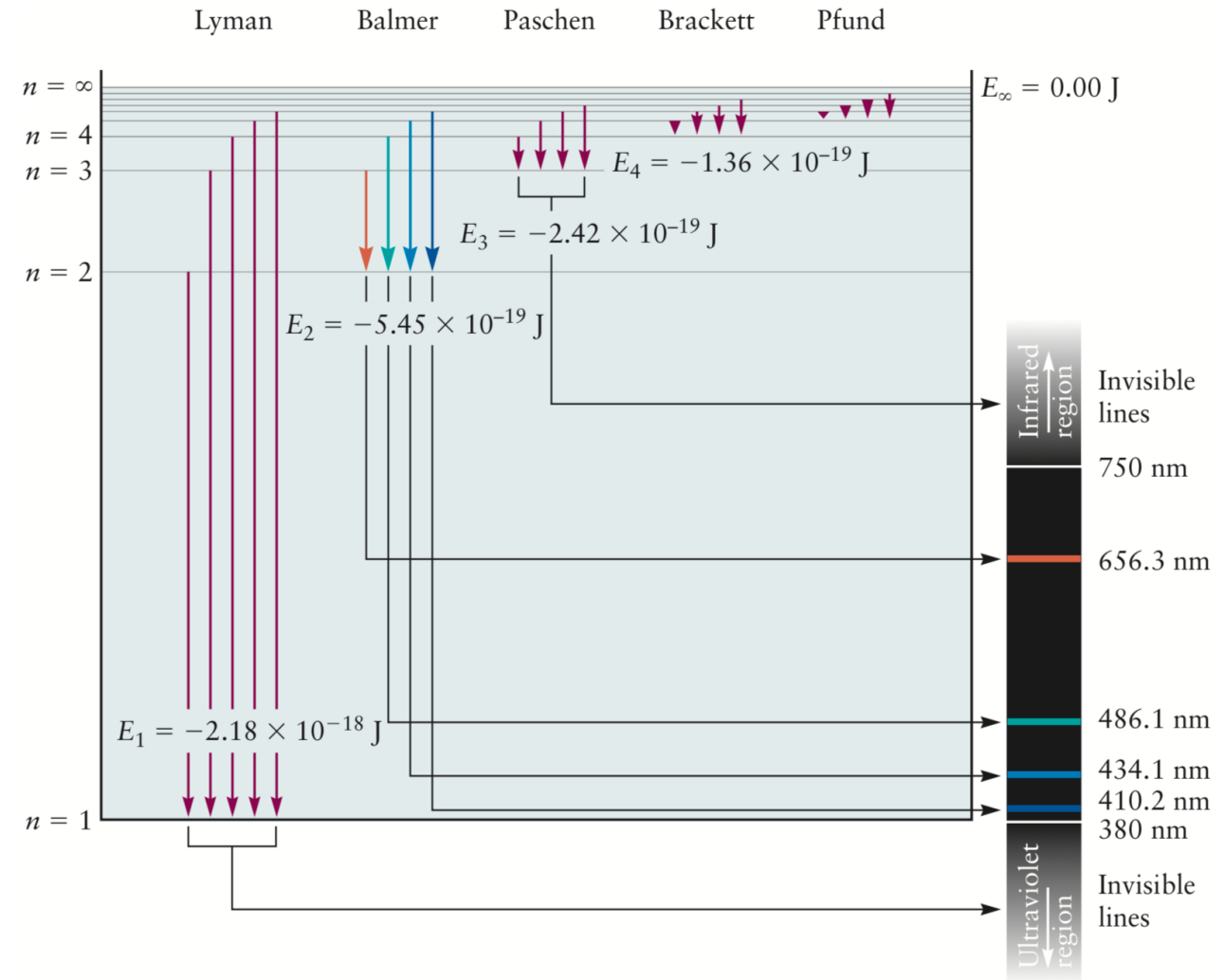
The Bohr model: predicting discrete energy levels in atoms

When a **one-electron atom or ion** undergoes a transition from a state characterized by quantum number n_i to a state lower in energy with quantum number n_f ($n_i > n_f$), light is emitted to carry off the energy $h\nu$ lost by the atom.

$$h\nu = \frac{Z^2 e^4 m_e}{8\epsilon_0^2 h^2} \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right] \quad (\text{emission})$$

$$\nu = \frac{Z^2 e^4 m_e}{8\epsilon_0^2 h^3} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = (3.29 \times 10^{15} \text{ s}^{-1}) Z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$n_i > n_f = 1, 2, 3, \dots (\text{emission})$$

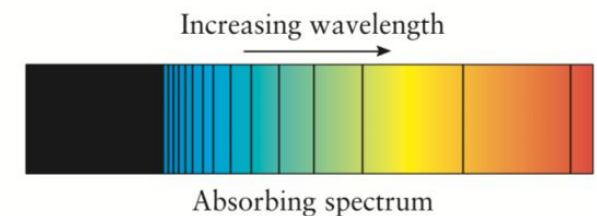
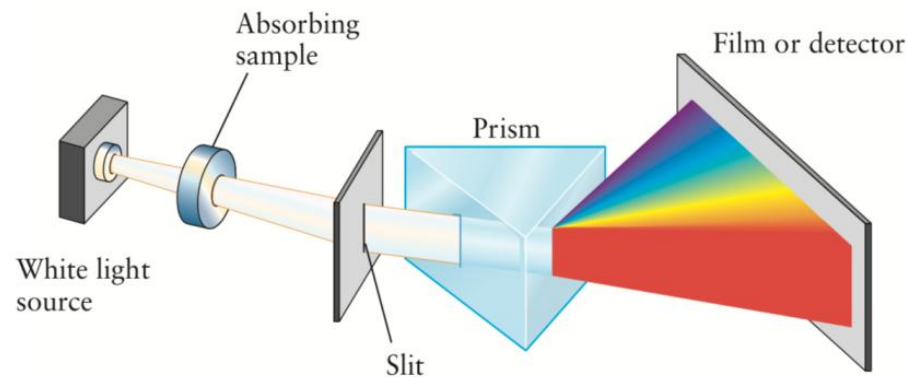
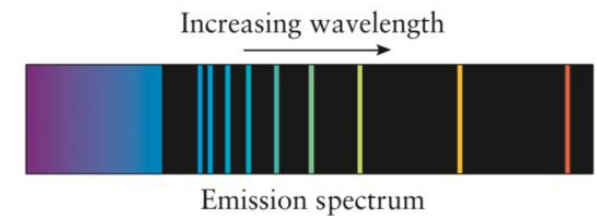
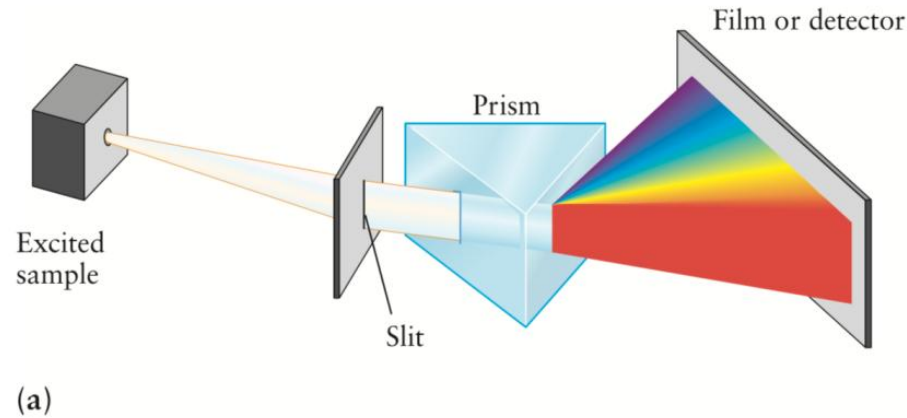


The Bohr model: predicting discrete energy levels in atoms

Conversely, an atom can absorb energy $h\nu$ from a photon as it undergoes a transition to a higher energy state ($n_f > n_i$). In this case, conservation of energy requires $E_i + h\nu = E_f$; thus, the absorption spectrum shows a series of lines at frequencies.

$$\nu = (3.29 \times 10^{15} \text{ s}^{-1}) Z^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$n_f > n_i = 1, 2, 3, \dots$ (absorption)



Atomic spectra

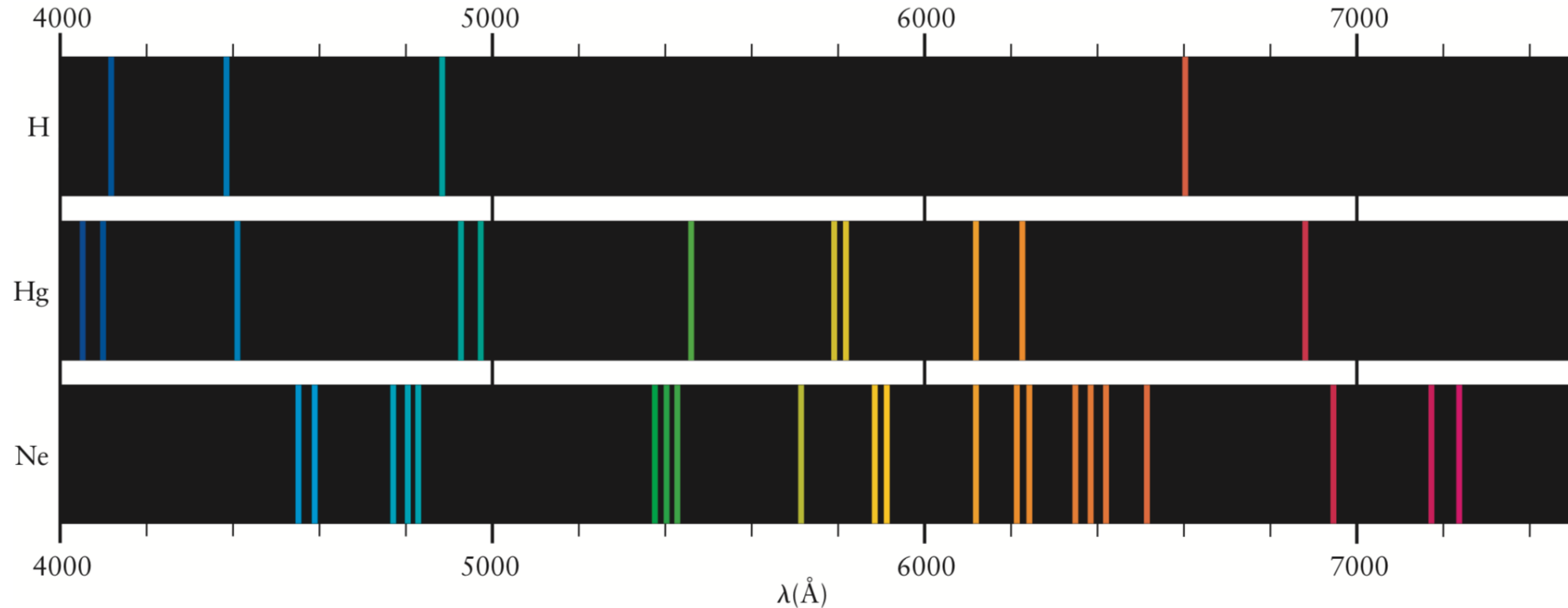
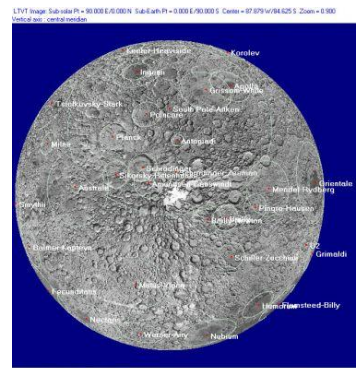


FIGURE 4.10 Atoms of hydrogen, mercury, and neon emit light at discrete wavelengths. The pattern seen is characteristic of the element under study. $1 \text{ \AA} = 10^{-10} \text{ m}$.

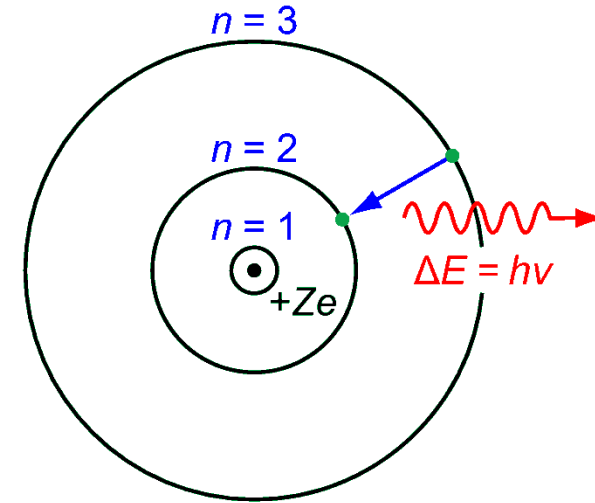
1885, J. J. Balmer discovered that hydrogen atoms emit a series of lines in the visible region, with frequencies given by the following simple formula:

$$\nu = \left[\frac{1}{4} - \frac{1}{n^2} \right] \times 3.29 \times 10^{15} \text{ s}^{-1} \quad n = 3, 4, 5, \dots$$



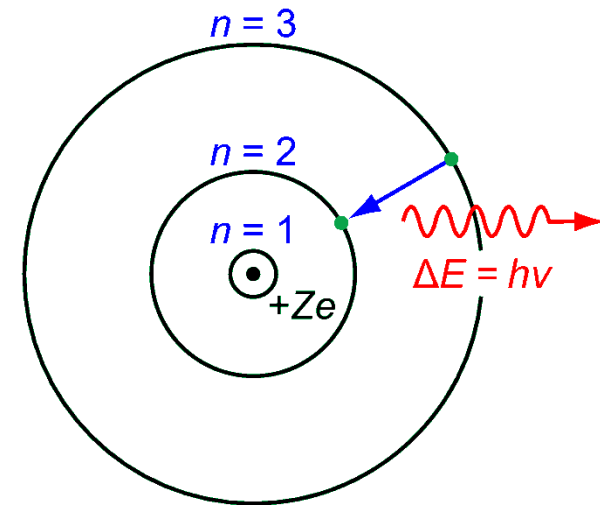
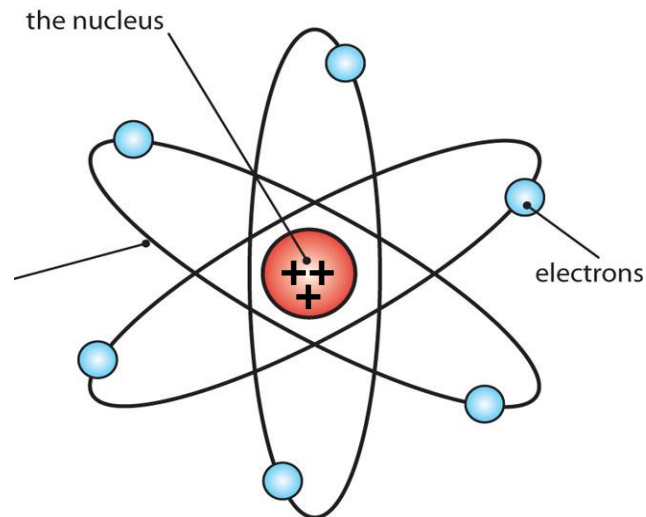
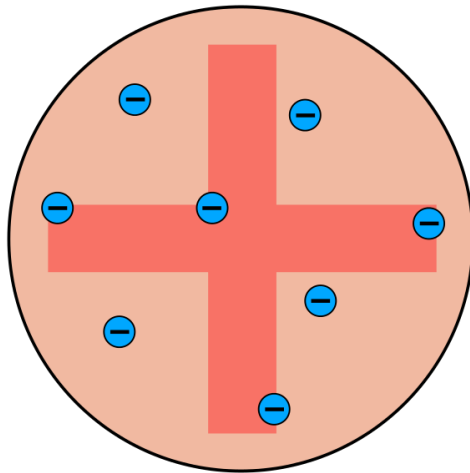
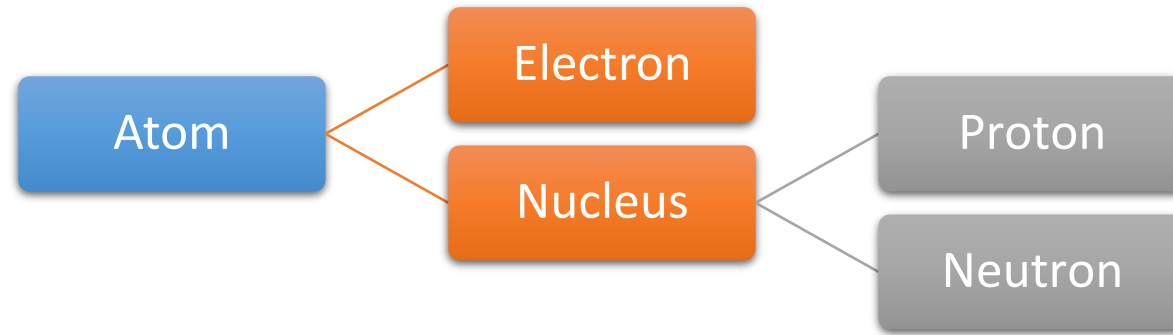
Summary

- Stable orbits / states of H atom
- Transitions between the orbits
- Quantized orbital angular momentum
- “Old quantum theory”



But the Bohr model doesn't work for the He atom at all!

Summary



Homework: read “principle of modern chemistry”, chapter 4

“Anyone who is not shocked by quantum theory has not understood it.”

Niels Bohr

